INCH-POUND

MIL-S-52881B
11 May 1989
SUPERSEDING
MIL-S-52881A(ME)
11 December 1985

MILITARY SPECIFICATION

SERVICING UNIT, COMPRESSED GAS, MOBILE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two types of mobile compressed gas servicing units (hereinafter called "servicing unit").

1.2 Classification. The servicing units shall be of the following types as specified (see 6.2):

Type I - Air
Type II - Dual air/oxygen

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

RR-C-901	- Cylinders, Compressed Gas, High Pressure, Steel DOT 3AA, and Aluminum Applications, General Specifications For.
UU-T-81	- Tags, Shipping and Stock.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3655
DISTRIBUTION STATEMENT A. Approved for public release, distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:11Z
Check the source to verify that this is the current version before use.

MILITARY

MIL-P-116	- Preservation, Methods of.
MIL-B-121	- Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-V-173	- Varnish, Moisture and FUngus Resistant (for Treatment of Communications, Electronic, and Associated Equipment).
MIL-P-514	- Plate, Identification, Instruction and Marking, Blank.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-B-7883	- Brazing of Steels, Copper, Copper Alloys, Nickel Alloys, Aluminum and Aluminum Alloys.
MIL-R-12144	- Reflector, Indicating, Clearance.
MIL-C-13486	- Cable, Special Purpose, Electrical, Low-Tension, Heavy Duty Single-Conductor and Multiconductor.
MIL-C-13486/12	- Cable, Special Purpose, Electrical, Low-Tension, Heavy Duty Multiple Conductors Unshielded.
MIL-D-16791	- Detergent, General Purpose (Liquid Non-Ionic).
MIL-G-20241	- Gasket Material, Wool Felt, Impregnated, Adhesive, Pressure-Sensitive.
MIL-T-22085	- Tape, Pressure-Sensitive, Adhesive, Preservation and Sealing.
MIL-S-40626	- Sign Kit, Vehicle Class.
MIL-R-46164	- Rust Proofing for Military Vehicles and Trailers.
MIL-C-46168	- Coating, Aliphatic Polyurethane, Chemical Agent Resistant.
MIL-B-46176	- Brake Fluid, Silicone, Automotive, All Weather Operational and Preservative.
MIL-C-62218	- Corrosion Prevention Compounds, Cold Application.
MIL-S-81733	- Sealing and Coating Compound, Corrosion Inhibitive.
MIL-C-83960	- Cylinder, Oxygen Purifier.

STANDARDS

FEDERAL

FED-STD-H28	- Screw Thread Standard for Federal Service.

MILITARY

MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-209	- Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.
MIL-STD-281	- Automobiles, Trucks, Truck-Tractors, Trailers and Trailer Dollies; Preservation and Packaging of.
MIL-STd-642	- Identification Marking of Combat and Tactical Transport Vehicles.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1179	- Lamp. Reflectors and Associated Signaling Equipment for Military Vehicles.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:11Z
Check the source to verify that this is the current version before use.

MIL-STD-1472	- Human Engineering Design Criteria for Military Systems, Equipment and Facilities.
MIL-STD-1791	- Design for Internal Aerial Delivery in Field Wing Aircraft.
MS35387	- Reflector, Indicating Clearance.
MS35423	- Light, Marker, Clearance and Service.
MS51331	- Guard, Splash, Wheel.
MS52125	- Composite Light - Tail, Stop, Turn and Marker.
MS52127	- Chock, Wheel.
MS75020	- Connector, Plug, Electrical - 12 Contact, Intervehicular, 24 Volt, Waterproof.

HANDBOOK

MILITARY

MIL-HDBK-113	- Guide for the Selection of LUbricants, Functional Fluids, Preservatives, and Specialty Products for use in Ground Equipment Systems.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATIN; NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

ME

13227E5951	- Valve, Ball, Quarter Turn.

(Copies of drawings required by contractors in connection with specific acquisition functions should be obtained from the USA Belvoir Research, Development, and Engineering Center, ATIN: STRBE- , Fort Belvoir, VA 22060-5606.)

THE MILITARY TRAFFIC MANAGEMENT COMMAND
Transportation Engineering Agency (MTMC/TEA)

Pamphlet 70-1 Transportability for Better Strategic Mobility.

(Application for copies should be addressed to the Military Traffic Management Command, Transportation Engineering Agency (MTMC/TEA), ATIN: MTT-TR, P.O. Box 6276, Newport News, VA 23606-0276.)

DEPARTMENT OF THE NAVY

NAVSEA 0910-LP-312-4600	- U.S. Navy Diving and Manned Hyperbaric Systems Safety Certification Manual.
NAVSEA 0994-LP-001-9010	- U.S. Navy Diving Manual, Volume 1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:11Z
Check the source to verify that this is the current version before use.

NAVSEA INST 10560.2 - Approved for Navy Use List.
Naval Experimental Diving Unit Instruction 7121A - Organizational Level Cleaning Procedures for Air and Oxygen Systems.

(Copies may be obtained from the Department of the Navy, Naval Sea Systems Command, Code OOC, Alexandria, VA 20362.)

DEPARTMENT OF LABOR (DOL)

29CFR1910 - Occupational Safety and Health Standards (OSHA).

(Application for copies should be addressed to the local DOL office or the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

DEPARTMENT OF TRANSPORTATION (DOT)

49CFR171-179 - Hazardous Materials Regulations.
49CFR390 and 393 - DOT Federal Motor Carrier Safety Regulations (FMCSR).
49CFR567 and 568 - Certification.
49CFR571 - DOT Federal Motor Vehicle Safety Standards (FMVSS).

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, 710 North Capitol Street, NW, Washington DC 20402.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

ASSOCIATION OF AMERICAN RAILROADS (AAR)

Rules Governing the Loading of Commodities on Open-Top Cars.

(Application for copies should be addressed to the Association of American Railroads, American Railroads Building, 50 F Street, NW, Washington, DC 20001.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI Z35.1 - Accident Prevention Signs, Specification for.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Section IX, Welding and Brazing Qualification.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:11Z
Check the source to verify that this is the current version before use.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM A 242 - Steel, Structural, High Strength Low Alloy.
ASTM A 449 - Bolt & Studs, Quenched & Tempered Steel.
ASTM E 527 - Numbering Metals and Alloys (UNS).

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia PA 19103.)

AMERICAN TRUCKING ASSOCIATIONS, INC. (ATA)

The Maintenance Council (TMC-ATA)

ATA Vehicle Maintenance Reporting Standards.
TMC-ATA Recommended Practice Manual.

(Application for copies should be addressed to the American Trucking Associations, 2200 Mill Road, Alexandria, VA 22314-4654.)

AMERICAN WELDING SOCIETY, INC. (AWS)

ANSI/AWS D 1.1 - Structural Welding Code, Steel.
ANSI/AWS D 1.2 - Structural Welding Code, Aluminum.

(Application for copies should be addressed to the American Welding Society, 550 N.W. LeJune Road, P.O. Box 351040, Miami, FL 33135.)

FRICTION MATERIALS STANDARDS INSTITUTE, INC. (FMS)

FMS Automotive Data Book.

(Copies may be obtained from commercial brake lining contractors or application for copies hsould be addressed to the Friction Materials Standards Institute, Inc., East 210, Route 4, Paramus, NJ 07652.)

INDUSTRIAL FASTENERS INSTITUTE (IFI)

IFI Standards.

(Application for copies hsould be addressed to the Industrial Fasteners Institute, 1505 East Ohio Building, 1717 East 9th Street, Cleveland, OH 44114.)

NATIONAL ASSOCIATION OF METAL NAME PLATE MANUFACTURERS, INC. (NAME)

Name Plate Standards and Practices.

(Application for copies should be addressed to teh National Assiciation of Metal Name Plate Manufacturers, Inc., 1700 East Dyer Road, Suite 165, Santa Anna, CA 92705.)

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:11Z
Check the source to verify that this is the current version before use.

NATIONAL WHEEL AND RIM ASSOCIATION (NWRA)

NWRA Catalog of Truck Wheels.
NWRA Cross Reference LIst.
NWRA Load Ratings for Truck Wheels.

(Application for copies should be addressed to the National Wheel and Rim Association, 4837 Victor Street, Jacksonville, FL 32207.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC (SAE)

SAE J163 - Low Tension Wiring and Cable Terminals and Splice Clips.
SAE J185 - Access Systems for Off-Road Machines.
SAE J410 - High Strength, Low Alloy Steel.
SAE J492 - Rivets and Riveting.
SAE J534 - Lubrication Fittings.
SAE J538 - Grounding of Storage Batteries.
SAE J553 - Circuit Breakers.
SAE J561 - Electrical Terminals - Eyelet and Spade Type.
SAE J592 - Clearance, Side Marker and Identification Lamps.
SAE J682 - Rear Wheel Splash and Stone Throw Protection.
SAE J683 - Tire Chain Clearance - Trucks, Buses (Except Suburbs, Intercity, and Transit Buses), and Combination of Vehicles.
SAE J684 - Trailer Couplings, Hitches and Safety Chains - Automotive Type.
SAE J697 - Safety Chain of Full Trailers or Converter Dollies.
SAE J749 - Drawbars - Crawler Tractor Industrial (Track Type) Tractor Equipment Mounting.
SAE J752 - Maintenance Interval - Construction Equipment.
SAE J753 - Lubrication Chart - Construction and Industrial Machinery.
SAE J754 - Lubricant Types - Construction and Industrial Machinery.
SAE J817 - Engineering Design Serviceability Guide Lines - Construction and Industrial Machinery.
SAE J847 - Trailer Tow Bar Eye and Pintle Hook/Coupler Performance.
SAE J849 - Connection and Accessory Locations for Towing Multiple Trailers.
SAE J875 - Trailer Axle Alignment.
SAE J925 - Minimum Service Access Dimensions for Off-Road Machines.
SAE J1047 - Tubing - Motor Vehicle Brake System Hydraulic.
SAE J1100 - Motor Vehicle Dimensions.
SAE J1292 - Automobile, Truck, Truck-Tractor, Trailer and Motor Coach Wiring.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

SOUTH COAST AIR QUALITY MANAGEMENT CONTROL DISTRICT

Rule 442 - Usage of Solvents.

(Application for copies should be addressed to the South Coast Air Quality Management Control District, 9150 East Flair Drive, El Monte, CA 91731.)

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:11Z
Check the source to verify that this is the current version before use.

TIRE AND RIM ASSOCIATION, INC. (TRA)

TRA Yearbook.

(Application for copies should be addressed to The Tire and Rim Association, Inc. 3200 West Market Street, Suite 304, Akron, OH 44313.)

TRUCK-TRAILER MANUFACTURERS' ASSOCIATION, INC.

TTMA Recommended Practices.
TTMA Technical Bulletins.

(Application for copies should be addressed to the Truck-Trailer Manufacturer's Association, Inc., 1020 Princess Street, Alexandria, VA 22314.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, (except for related associated detail specifications, specification sheets or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The servicing units shall be compatible for the resupply of breathable, life-supporting compressed gas with self-contained underwater breathing apparatus (SCUBA) cylinders for divers. Type I servicing units shall be for breathing air use only and shall contain the components shown in figures 2 and 3. Type II servicing units shall be for dual breathing air and oxygen use, shall contain the components and be divided as shown in figures 4 and 5. Both servicing units shall be compatible with military trucks for towing over highways and off-road terrain between the compressed gas source and the site of diving operations.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection.

3.3 Materials. Materials shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Material deterioration prevention and control. The servicing unit shall be fabricated from compatible materials, inherently corrosion and deterioration resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable storage and operating environments to which the servicing unit may be exposed. Corrosion-resisting treatment involving galvanizing and electrodepositing of chromium, copper, or nickel shall not be furnished on any life support component of the servicing unit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:11Z
Check the source to verify that this is the current version before use.

3.3.1.1 Dissimilar metals. Dissimilar metals, as defined in MIL-STD-889 and in NAVSEA 0910-LP-312-4600, shall be electrically insulated from one another to minimize or prevent galvanic corrosion. Insulation may be provided by an insulating barrier such as a corrosion-inhibiting sealant conforming to MIL-S-81733 or chromate tape conforming to MIL-G-20241. Protection against corrosion could also be obtained by exclusion of the electrolyte, if feasible.

3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.3.2 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the servicing unit may be newly fabricated from recovered materials to the maximum extent practicable, provided the servicing unit produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the servicing unit.

3.4 Maintainability. The servicing units shall be accessible for servicing, repair, and replacement. The adjustment and replacement of components shall be accomplished with minimum disturbance to adjacent components. Where incorrect installation of an item could cause malfunction of that item or of the system in which it is installed, an unsymmetrical mounting means shall be provided or the mating parts shall be doweled or matchmarked. Ease of maintenance provisions shall incorporate features insuring operating clearances in accordance with SAE J925 and facilitate maintenance and service operations. The servicing unit shall be maintainable using only those common mechanic's tools. The servicing units Maintainability Index and Repairability Index, in accordance with SAE J817, shall be computed and included in the manual. Test in accordance with 4.6.2.4. Except as stated above, design for maintainability shall be in accordance with 3.7.

3.4.1 Maintenance support. Assemblages or support elements such as technical manuals, repair parts, special tools, or lubrication charts, as specified (see 6.5 and 6.8) shall be adequate to perform the intended function of assisting or conducting maintenance operations when tested as specified in 4.6.2.4, and shall not require scheduled maintenance more often than every 100 hours of operation.

3.4.2 Maintenance ratio. The servicing units shall have a maintenance ratio of not more than 0.04 when tested as specified in 4.6.2.4. Maintenance ratio is defined as the ratio of the total active maintenance man-hours required (scheduled and unscheduled) to the total operating time. Man-hours for repair of replaced components and scheduled before-and-after operation checks are excluded. A maintenance schedule shall be furnished prior to the start of any testing.

3.5 Design. The servicing unit shall consist of compressed gas cylinders installed in a rack-type body assembly. The body assembly, in turn, shall be mounted on a trailer assembly as specified herein. The servicing unit shall provide a complete servicing system when connected to underwater diving (SCUBA) cylinders. Components and subassemblies utilized in the air/oxygen systems shall be obtained from approved sources as listed in NAVSEA INST 10560.2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:11Z
Check the source to verify that this is the current version before use.

3.5.1 Body assembly. The body assembly shall secure and support eight DOT 3AA standard cylinders, conforming to RR-C-901, in the horizontal position (see figure 1). The cylinders shall be capable of being slid in or out from the rear of the body assembly and, when installed, shall be secured in place against a front bulkhead. A gas flow control system shall be furnished, including manifolds, fittings, piping, valves, gages, safety devices, purifier, and a control panel, all located forward of the bulkhead (see figures 2 and 4 for flow diagrams).

3.5.2 Trailer assembly. The trailer assembly shall be as specified in 3.10.6.

3.6 Transportability. The servicing units shall be capable of being transported by military or commercial trailers, trains, marine vessels, aircraft and must be capable of withstanding the impact forces encountered in shipment without damage or permanent deformation. The servicing units shall be equipped with tiedown or slinging provisions.

3.6.1 Tiedown provisions. The tiedown provisions shall conform to MIL-STD-209, class 2 or 3, type II equipment, and to MIL-STD-1791 for equipment restraint criteria. The tiedown provisions shall satisfactorily complete the pull testing as specified without weld failure, permanent deformation, cracking, loosening, or breaking of the provision or its connecting structural components.

3.6.2 Slinging provisions. The slinging provisions shall conform to MIL-STD-209, class 2 or 3, type II. The provisions shall enable the servicing units to be lifted in the normal operating position. The provisions shall be located so that not less than one inch clearance is maintained between slings and all exterior parts and shall be fastened to members which will withstand stresses in the amount and direction of pull specified for the provisions without weld failure, permanent deformation, cracking, loosening, or breaking of the provision or its connecting structural components. No spreader bars are permitted. Slinging provisions may also be used as tiedown provisions when such provisions meet the requirements specified in 3.6.1. All slinging tiedown provisions shall be labeled 'LIFT', 'TIEDOWN', as appropriate, in 1-inch (2.54 cm) high letters.

3.6.3 Air transportability. The servicing units shall meet the requirements of MIL-STD-1791 for air transport.

3.6.4 Rail transportability. The servicing units shall be rail transportable in CONUS and NATO countries without restrictions. The servicing unit shall have a dimensional profile within the Gabarit International de Chargement (GIC) in accordance with MTMC/TEA pamphlet 70-1, outline diagram when loaded on a 50-inch (127 cm) high rail car. The servicing units shall be capable of withstanding shock loads resulting from rail impact testing in accordance with 4.6.2.6.1.2.3 without failure, damage, or permanent deformation. After testing, the servicing units shall operate as specified herein for a period of not less than 1 hour without malfunction.

3.6.5 Highway transportability. The servicing units when loaded on a semitrailer/tractor, shall be within the highway permit limits of all States.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:11Z
Check the source to verify that this is the current version before use.

3.7 Human factors engineering. The human factors engineering characteristics of the servicing unit shall meet the following minimum requirements:

a. Conform to the design criteria of MIL-STD-1472. Special design emphasis shall be given, but not limited, to the following: 4. General Requirements; 5.1 Control/Display Integration; 5.2 Visual Displays; 5.4 Controls; 5.5 Labeling; 5.6 Anthropometry; 5.9 Design for Maintainability; and 5.13 Hazards and Safety.
b. Be operable and maintainable by personnel ranging (1) from the 5th percentile through the 95th percentile male as described in MIL-STD-1472, 5b.
c. Provide all walk, step, and hand-hold components per SAE J185.

3.8 Safety. The servicing units shall conform to the following safety requirements:

a. Comply with DOL OSHA Standards (26CFR1910).
b. Comply with the "Hazardous Materials Regulations" of the Department of Transportation (DOT), 49CFR177, including placards.
c. Conform to DOT Federal Motor Vehicle Safety Standards (FMVSS) 49CFR567 and 49CFR571, and DOT Federal Motor Carrier Safety Regulations (FMCSR) 49CFR390.
d. Result in furnishing material and components having a correct safety factor.
e. Result in furnishing material and components having a correct safety factor.
f. Eliminate nonfunctional sharp edges, projecting points, and excessive lengths of fasteners.
g. Provide, as applicable, standard safety color codes for marking of the vehicle's physical hazards in accordance with ANSI Z35.1.

3.9 Environmental requirements.

3.9.1 Operational temperatures. The servicing unit shall not be damaged during operation and transport in any ambient air temperature between -25 °F and +120 °F. Test for operational temperatures as specified in 4.6.2.8.

3.9.2 Storage temperature. The servicing units shall not be damaged by storage in any ambient temperature from +155 °F for 4 hours daily to -65 °F without solar radiation for 12 hours (see 4.6.2.8).

3.9.3 Precipitation. The servicing unit shall not be damaged or become inoperative when exposed to rain ranging from a drizzle to a heavy rain of 1/4 gallon per minute per square foot of wetted surface impinging on the components of the servicing unit with a 30 mph wind. During and after the precipitation, the servicing unit shall operate as specified herein (see 4.6.2.8).

3.10 Servicing unit construction. The servicing unit shall be constructed of a body assembly bolted onto a trailer chassis as specified herein.

3.10.1 Body assembly. The body assembly shall be fabricated of aluminum structural framing with exterior panels of either aluminum trailer sheeting or glass-reinforced plastic trailer sheeting. Body panels shall be readily removable or shall open to facilitate maintenance of the components, especially

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:11Z
Check the source to verify that this is the current version before use.

between the control panel and the bulkhead. An underbody shield shall be furnished to protect the servicing unit components from road splash and gravel damage. The control panel shall be protected by cabinet doors. The doors shall include paddle-type slam latches with strikers. The hinges shall be of the piano type having minimum 1/4-inch diameter brass or stainless steel hinge pins. Catches shall be furnished to hold the doors open during windy weather. A padlock hasp shall be furnished to lock the doors. The hasp shall accept a padlock having a minimum 3/8-inch diameter shackle. A plastic document holder shall be furnished on the inside of the curbside control panel door (see 3.14.3.2). At least two each drain holes with elastomer plugs shall be provided in the bottom of the body assembly.

3.10.2 Cylinder rack. The cylinder rack shall accommodate eight cylinders lengthwise on the trailer-mounted unit as shown in figure 1. There shall be individual troughs and adequate supporting framework with provisions to prevent the accumulation of water around the cylinders when the trailer is in its normal position. Each cylinder shall be removable separately without moving the others and shall have plastic strips the length of the cylinder to prevent damage from vibration. The rack, with bulkhead and clamping, shall securely hold the compressed gas cylinders against movement during the operation and mobility tests specified herein, including railroad car humping.

3.10.2.1 Clamps. A quick-acting and easily accessible clamping arrangement, designed to hold the cylinders in place, shall be provided integral with the cylinder rack. Clamps shall not require tools to operate.

3.10.2.2 Bulkhead. A cylinder bulkhead with openings for cylinder necks shall be provided at the front of the cylinder rack.

3.10.3 Gas flow control system. The component piping, fittings, valves, gages, purifier(s), regulator(s), and connecting lines of the air flow control system shall be designed for a working pressure of not less than 5000 pounds per square inch gage (psig) and 3000 for the oxygen flow control systems. They both shall be of ASTM S30400 stainless steel in accordance with ASTM E 527, suitable for the purpose, cleaned, and shall be compatible for the gas service intended. Safety devices shall be provided at each manifold, at each regulator, and as required to properly protect the compressed gas system. The gas flow control system shall be capable of withstanding a hydrostatic pressure of not less than five-thirds (5/3) of the working pressure. All components of the gas flow control system shall be delivered in an oxygen-clean state as specified in 3.14.3. Test in accordance with 4.6.2.5.

3.10.3.1 Fittings. Gas fittings on each of the two types of gas flow control systems shall be in accordance with FED-STD-H28 as follows: Connection No. 540 for oxygen service and connection No. 347 for air service. All fittings shall be of ASTM S30400 material.

3.10.3.2 Manifold system. The cylinders shall be manifolded as shown in figure 2 for the type I servicing unit and as shown in figure 4 for the type II servicing unit. Each cylinder shall be connected to a pressure gage and a quarter-turn ball valve with a flexible wire-braided hose. All manifold pipe connections when not threaded into the associated valves, shall be welded connection/unions in accordance with 3.20.3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:11Z
Check the source to verify that this is the current version before use.

3.10.3.2.1 Flexible connecting lines. Flexible wire-braided hoses for high pressure gas service shall be furnished as connecting lines from the storage cylinders to the manifold. Fittings at each end shall be as specified in 3.10.3.1. All connecting lines shall be of ASTM S30400 material.

3.10.3.2.2 Individual pressure gages. Individual moistureproof pressure gages shall be provided for each cylinder. The gages for the air bottles shall have a pressure range of 0 to not less than 6000 pounds per square inch (psi) in 100 psi increments and shall be not less than 2-1/2 inches in size. The gages for the oxygen bottles shall have a pressure rating range of 0 to not less than 3000 psi in 100 psi increments and shall not be less than 2-1/2 inch in size. Cylinder-side fittings shall be as specified in 3.10.3.1 to mate with the flexible lead fittings. Gages for oxygen service shall be clearly labeled. The oxygen gages shall not be interchangeable with the gages on the air side of type II servicing unit panels.

3.10.3.2.3 Valves. Manually operated shutoff, isolation, and control service valves shall be of the positive shutoff type and shall be of ASTM S30400 material. The valves shall require not more than 60 inch-pounds of torque to open or close under a pressure of 5000 psi and overtorquing to 150 inch-pounds shall not damage the valve seats. Valve seats shall be resistant to corrosion and heat damage.

3.10.3.2.4 Quarter-turn ball valves. Valves shown in figures 2 and 4 as quarter-turn ball valves shall be 2-way, straight-through design quarter-turn ball valves. The valves shall conform to the dimensional limits shown on 13227E5951. One valve shall be provided for each cylinder. Each valve shall provide positive shutoff to working pressure (3000 psi) for oxygen systems and (5000 psi) for air systems from the manifold side and the cylinder side of the valve. The piping connections shall be 1/4-inch female NPT in the arrangement shown on 13227E5951. Seats in the valves shall be polytetrafluoroethene. The valve body shall be ASTM S30400 material. Handles shall be sized and spaced in accordance with MIL-STD-1472, rotary controls. The "On" and "Off" positions shall be spaced 90 degrees apart and they shall be clearly marked in accordance with 3.7. The valve handle shall point up to indicate the "On" position.

3.10.3.2.5 Recharging valve. A recharging valve shall be incorporated in the manifold system to provide facility for recharging the storage cylinders without removing them from the servicing unit. The valve shall have a design working pressure of 5000 psig and 3000 psig for oxygen systems and an outlet connection as specified in 3.10.3.1.

3.10.3.3 Purifier. A cartridge-holding-type MB-1 cylinder conforming to MIL-C-83960 shall be incorporated as shown in figure 2 for type I servicing units and as shown in figure 4 for type II servicing units.

3.10.3.4 Manifold shutoff valve. A manifold shutoff valve shall be located between the manifold and the purifier cylinder as shown in figure 2 for type I servicing units and as shown in figure 4 for type II servicing units.

3.10.3.5 Regulators. Regulators shall be provided. The regulators shall be designed for a normal working pressure of 5000 psig and shall be adjustable to any outlet pressure setting between 50 psig and a maximum of not less than 3000 psig, inclusive. Each regulator shall incorporate features which protect it and

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:11Z
Check the source to verify that this is the current version before use.

its pressure control capability against inlet pressure surges. The regulator shall allow gas flow in only one direction so as to prevent backflow when pressure in the SCUBA charging manifold is greater than that in the supply cylinder manifold. The regulator shall not emit gas to the atmosphere by such features as back-venting or self-venting. Each regulator shall have an inlet and an outlet pressure gage.

3.10.3.6 Inlet pressure gage. A inlet pressure gage shall be provided to indicate regulator inlet pressure. The gage shall have a range scale of 0 to not less than 6000 psi for air systems and 300 psi for oxygen systems with 100 psi increments. The dial size shall be not less than 4 inches. One gage as shown in figure 2 for type I servicing units and two gages as shown in figure 4 for type II servicing units shall be provided.

3.10.3.7 Outlet pressure gage. An outlet pressure gage shall be provided to indicate the regulator outlet pressure. The gage shall have a scale range of 0 to not less than 6000 psig for air systems and 3000 psig for oxygen systems with 100 psig increments. The dial size shall be not less than 4 inches. One gage as shown in figure 2 for type I servicing units and two gages as shown in figure 4 for type II servicing units shall be provided.

3.10.3.8 SCUBA charging manifold. A manifold as shown in figure 3 shall be provided for simultaneous charging of four SCUBA cylinders for type I servicing units. Two manifolds as shown in figure 4 shall be provided for simultaneous charging of four SCUBA cylinders (two with air, two with oxygen) for type II servicing units. For type I servicing units, each manifold shall have a bleed-off valve. Wire-braided ASTM S30400 material, polytetraflouroethene lined hose charging leads (not less than 15 feet long) and yoke-type filling adapters shall be provided for connecting to standard "J" and "K" type SCUBA cylinder valves. The yoke adapters shall be of the type having integral bleed-off valves. For type II servicing units, the oxygen manifold shall have a bleed-off valve. Flexible wire-braided ASTM S30400 material, polytetraflouroethene lined hose charging leads (not less than 15 feet long) and yoke-type filling adapters shall be provided for connecting to standard medical port outlet in accordance with FED-STD-H28 (connection No. 870). The yoke adapters shall be of the type having integral bleed-off valves of ASTM S30400 material.

3.10.3.8.1 SCUBA cylinder trough. A trough shall be provided to hold water for cooling of SCUBA cylinders during charging. The trough shall be constructed of corrosion-resistant materials or materials which are treated for corrosion resistance. A drain outlet with plug shall be included. The trough shall be a minimum of 31 inches in depth and shall be a minimum of 12 inches wide by 40 inches long. The trough may be constructed in two sections. Provisions shall be made for mounting the trough on the servicing unit.

3.10.3.9 Control panel. A control panel shall be mounted forward of the storage cylinders at a distance sufficient to provide accessibility for operation and maintenance of the installed equipment. The valves, gages, manifold, regulator(s), purifier(s), and connecting lines shall be securely mounted on the panel. All components mounted on the panel shall be positioned so as not to interfere with or obstruct the use or removal of adjacent items. No component or portion thereof shall extend beyond the top or outer edges of the

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:11Z
Check the source to verify that this is the current version before use.

panel. The components of the dual gas system (type II) shall be mounted on separate halves of the control panel as shown in figure 5. The control panel design shall be in accordance with 3.7.

3.10.3.10 Cylinders. The cylinders if not government furnished shall have at least 48 months remaining on each of the cylinders hydrostatic test period. The cylinders shall be approved standard DOT 3AA cylinders with the nominal size being 51 inches to 55 inches high by 9 inches in diameter. Each cylinder shall have a minimum working pressure of 5000 psi for air and 3000 psi for oxygen.

3.10.4 Servicing unit toolbox. A metal weather-resistant toolbox for stowing only the compressed gas servicing unit repair parts and maintenance and operating tools shall be provided in or on the servicing unit body assembly. The toolbox shall be identified with a warning sign (see 3.15.3) to read, "Toolbox and tools are to be used on life support equipment only. Do not contaminate with any type of oil or grease (hydrocarbon-POL) products." The warning shall be in accordance with Department of the Navy's NAVSEA 0994-LP-001-9010. The toolbox shall be accessible through a top lid or side door lid of the drop-down type. The lid shall be secured by a continuous hinge having a minimum 1/4-inch diameter brass or stainless steel pin extending the full length of the lid. Two tension-type fasteners with strikers shall be provided to retain the lid in the closed position. A hasp shall be fastened to the lid to lock the toolbox. The hasp shall accommodate a padlock having a minimum 3/8-inch shackle. The fasteners, strikers, and hasp shall be attached to the toolbox by either bolts or rivets. When closed, the lid shall prevent rain and dirt from entering the toolbox. The lid shall open through an arc of at least 90 degrees and shall remain open by its own weight or by holding devices. At least two each drain holes with elastomer plugs shall be provided in the bottom of the toolbox.

3.10.5 Servicing unit body mounting. The servicing unit body and equipment (see 3.10 through 3.10.4) shall be mounted wit removable fasteners to the trailer in 3.10.6 in such a manner that the servicing unit shall remain operational when separated from its trailer. The servicing unit body shall mount on the trailer chassis similar to that shown in figures 6 and 7. The body, equipment, and chassis match shall:

a. Provide a mounting covering a minimum of three each chassis frame crossmembers to provide a solid support for the body mounts as shown in figures 6 and 7.
b. Achieve the weight on the front tow eye of the loaded trailer chassis within the allowable percent of gross vehicle weight rating (GVWR) range specified in table II (see 3.12). Correctly locate the horizontal center of gravity of the servicing unit body forward of the center of the trailer chassis running gear with the compressed gas cylinders both full of gas and empty.
c. Provide twin bolted-on, side shear brackets and fasteners to position and secure the servicing unit body in accordance with figure 7.
d. Insure that the fenders and similar parts that extend below the servicing unit body do not interfere when the servicing unit body is removed from its trailer and set on the ground, on a platform, or on a boat deck.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:11Z
Check the source to verify that this is the current version before use.

3.10.6 Trailer specifications. The trailer specifications shall be in accordance with table I. The component sequence and number system shall be in accordance with the "Vehicle Maintenance Reporting Standards: (VMRS), the trucking industry standardization system of the American Trucking Association (ATA).

TABLE I. Trailer specifications.

VMRS No.	Feature/Requirements
Group 10	Chassis. - Provide strength, flexibility, and rigidity to withstand dynamic transport, both highway and off-road. - Withstand operational site jacking and shock and vibration imposed by the operating equipment and accessories. - Provide a firm foundation for the operating equipment insuring correct alignment of machinery and distribution of weight without incurring abnormal stress concentrations; verify by stress analysis (see 4.6.2.2). - Provide components that are bolted on in lieu of welded on to the most practical commercial extent.
VMRS 12-001	Axle, Rear, Non-driven. - Provide single axle having minimum 5000 lb gross axle weight rating (GAWR) at the axle hubs. - Rate axles per DOT FMVSS and TTMA Recommended Practice (RP) No. 39 and each axle manufacturer's maximum. Minimum axle ratings to include the bearing and beam rating, the brake rating, and the resulting partial GAW at the hub (without tire and wheel). - Provide running gear for floatation as specified. - Provide spindles with hardened surfacing for bearings and seals.
VMRS 13	Brakes. - Provide brake system per DOT FMVSS, DOT FMCSR. - Provide service, emergency-breakaway, and parking braking capability. - Provide brakes of the drawbar-actuated, surge type, hydraulic brakes suitable for tow vehicles without air available for brakes. Disc or drum brakes are acceptable.
VMRS 13-001-026	Disk Brake. - Provide rotors of the high heat sink type incorporating heavy use, cooling feature design. - Provide rotors that are warp resistant when overheating occurs. - Provide weatherproof puck seals which effectively seal between the puck and cast housing against malfunctions caused by road salt splash, dust, mud, water, snow, and ice.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:11Z
Check the source to verify that this is the current version before use.

TABLE I. Trailer specifications - continued.

VMRS No.	Feature/Requirements
VMRS 13-002	Rear Brakes and Drums. - Protect brake components located below the axle housing and inside, between the backing plates; however, locate no components below the bottom of the brake rotor or brake drum. - Provide components between the drum and the backing plate or dust shields either cleaned, treated, and painted or inherently corrosion resistant. - Provide dust shield protection of foundation brake assemblies. - Apply a thin wipe of silicone high temperature compound on brake system bearing surfaces.
VMRS 13-002-014	Lining, Rear Brake. - Lining shall be selected from the FMS Automotive Data Book. - Lining shall be rated per DOT procedures and technically intended for severe, mountainous, heavy duty commercial service. - For disc type brakes, provide linings of the semi-metallic or full-metallic type.
VMRS 13-002-023	Drum, Rear Brake. - Provide brake drums of the heaviest regular production optional weight available.
VMRS 13-006	Brake Master Cylinder. - Provide surge brake system master cylinder actuator and installation to comply with structure design, loadings, and safety of SAE J684, except substitute an SAE J847 tow eye (see VMRS 59-007) instead of ball and socket type illustrated. - Provide surge brake actuators having a minimum 6000 lb gross load rating. - Fill all hydraulic master cylinder systems with DOT-5 brake fluid (VMRS 53-999-008) meeting the minimum requirements of either FMVSS No. 116, DOT-5, or MIL-B-46176 for diorgano polysiloxane fluid (silicones). - Provide label on master cylinder for SBF or DOT-5.
VMRS 13-006-008	Plug, Master Cylinder Filler. - Provide means to prevent water (during creek bed traverse) from being sucked into the master cylinder reservoir through the filler plug breathers. - Provide inverted U-tube breathers, having at least 6 inches of air hose on the open end, aiming downward, to trap air.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:11Z
Check the source to verify that this is the current version before use.

TABLE I. Trailer specifications - continued.

VMRS No.	Feature/Requirements
VMRS 13-007	Brake Lines. - Hydraulic brake lines shall be in accordance with DOT FMVSS for flex lines and SAE J1057 for steel tubing lines. - Provide flex hose with swivel fittings furnished between master cylinder and frame and between frame member or the suspension or axle. - Provide routing of the lines within the protection of the trailer framing members.
VMRS 16	Suspension. - Provide single axle, flexibly sprung, suspension system, tailored to the load imposed and ranging between 70 to 100 percent of the axle rating, as necessary for a nondamaging ride. - Provide double-acting telescopic heavy duty shock absorbers. - For off-road travel over adverse terrain, provide stabilized ride of the trailer without hazardous handling. - Provide soft, nondamaging ride on highway. - Provide suspension to protect trailer from damaging shock and vibration. - Provide transverse roll angle stiffness such that a maximum 7.5 degree angle, between axle centerline and trailer frame, is not exceeded at GVWR while traversing a level roadway curve producing up to 0.5 g's lateral acceleration. - Provide positive control over axle rotation and tendency of the axles to brake-hop during braking. - Provide suspension having minimum 4 inches of travel articulation from full compression to rebound. - Provide elastomer stops and bumpers to prevent metal-to-metal contact and damage when the axle reaches maximum wheel compression position and when the axle reaches maximum wheel rebound position. - Provide suspension installation in proper alignment with tow eye and trailer centerline, per both suspension and axle manufacturers' recommendations and in accordance with SAE J875 and TIMA RP No. 71. - Spring working loads shall be rated in accordance with TIMA RP No. 32.
VMRS 17	Tires, Tubes, Liners and Valves.
VMRS 17-001	Tire, Pneumatic. - Tires furnished to the Government shall not be more than 12 months old from the date of cure of the rubber to date of acceptance by the Government. - Tires shall comply wit TRA Yearbook and DOT FMVSS for highway speeds.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:11Z
Check the source to verify that this is the current version before use.

TABLE I. Trailer specifications - continued.

VMRS No.	Feature/Requirements
VMRS 17-001 (continued)	Tire, Pneumatic. - Tires shall be at least 100 level quality and intended for use on powered units. - Provide all tires with traction-road-lug type tread. - Provide 3 each wide-base, single, tubeless tires, minimum size 12R 16.5, load range E, having maximum 35 psi tire inflation pressure for flotation use.
VMRS 17-008	Valve Assembly. - Provide metal valve caps with elastomer seals for all tire assemblies.
VMRS 18	Wheels, Rims, Hubs and Bearings. - Provide 3 each tire and wheel assemblies; 2 each on the axle plus 1 each spare.
VMRS 18-003-001	Wheel Assembly (Rim & Wheel), Disc. - Disc wheels shall be in accordance with DOT FMVSS, TRA Yearbook, and NWRA specifications. - Provide 3 each wheels per NWRA Wheel No. X-45249, rim size 16.5 x 9.75, 8-hole on 6.5-inch diameter bolt circle. - Provide tire and wheel assemblies inherently in balance.
VMRS 18-004	Trailer Hubs and Bearings. Provide axle hubs for disc wheels. - Studs and bolt circle match to wheels in VMRS 18-003-001. - Standardized tapered roller bearings with cups, bearings, and cones, all matched in accordance with the bearing and axle manufacturer's recommendations. - Bearings per TMC-ATA Advisory AV6-15.
VMRS 18-004-011	Seal, Oil, Wheel. - Provide lube oil seals per TTMA RP No. 43.
VMRS 18-004-017	Dust Cap, Wheel Hub. - Provide visual dust caps for oil lube level.
VMRS 30	Electrical Group. - Provide a complete military service and blackout vehicle electrical system per DOT FMVSS, DOT FMCSR, and MIL-STD-1179.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:11Z
Check the source to verify that this is the current version before use.

TABLE I. Trailer specifications - continued.

VMRS No.	Feature/Requirements
VMRS 30 (continued)	Electrical Group. - Provide military 24-volt, dc system with negative ground per SAE J538.
VMRS 34-002	Lamps, Rear, Tail, Stop, Turn, License. - Provide 2 each composite lights on the rear for both service and blackout tail, stop, turn, and clearance per MS52125, 24 volt. - Install lamps as far apart as practical in the rear bumper frame member or in lamp protectors above the rear bumper; lamp protectors at least equal to Schedule 40, 8- or 10-inch diameter steel pipe. - Recess lamps forward from the rearmost edge of the trailer to provide lamp protection when backing.
VMRS 34-003-088	Plug (12 Conductor Electrical Connector) Military 24 Volt System (MS75020-1) Pin Type.
VMRS 34-003-092	Connecting Cable, Vehicle to Vehicle, 12 Wire Military, 24 Volt (MIL-C-1348612). - Provide front intervehicular plug and cable assembly for military truck towing. - Furnish a 12-way plug per MS75020-1 connected to a cable manufactured in accordance with MIL-C-13486/12. - Permanently connect plug and cable assembly to trailer chassis to prevent loss. - Plug to extend a minimum of 30 inches toward the truck from the tow bar eye center. - Provide clip to hold the plug with the cable doubled back on the tow bar for stowage. - Locate plug clip to minimize jackknife damage when truck and trailer maneuver with cable stowed.
VMRS 34-004	Wiring, Fuses, Lamps, Bulbs. - All wiring per SAE J163 and SAE J1292 using minimum 14 AWG wire per MIL-C-13486. - Wire identification codes installed on each end of each wire for all circuits; see VMRS 34-004-031, Tabulation A. - Combine multiple circuits into harnesses or encase in thin wall conduit. - Wiring supported at maximum 36-inch spacing with anchored elastomer ties or coated clamps to prevent whipping, chafing, and mechanical or fatigue damage. - Interior of mechanical splicing devices waterproofed with RTV silicone sealant. - Wire equipped with terminals per SAE J561 or military type.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:11Z
Check the source to verify that this is the current version before use.

TABLE I. Trailer specifications - continued.

VMRS No.	Feature/Requirements
VMRS 34-004 (continued)	Wiring, Fuses, Lamps, Bulbs. - Disconnect points for multiple circuits (in excess of 4 each) of the waterproof plug type or of the nut and stud junction block type. - Junction blocks coated with electrical varnish after assembly. - Wire-to-wire terminals secured to vehicle. - For return ground, provide ground wires between the front receptacle, body, equipment, and chassis framing members. - On a metal plate or decal, provide a vehicular wiring diagram, as actually furnished on the trailer, similar to TIMA Technical Bulletin No. 65.
VMRS 34-004-010	Breaker, Circuit. - Provide circuit breakers per SAE J553 protecting all live circuits between the trailer plug and the lamps. - No fuses acceptable.
VMRS 34-004-025	Reflector, Amber.
VMRS 34-004-0026	Reflector, Red (Side). - Furnish reflectors per MIL-R-12144, MS35387, and TIMA RP No. 9, plus 2 each additional on the front facing forward. - Secure all reflectors to the vehicle with metal fasteners. - No reflectors installed on the unprotected trailer surfaces or at an angle greater than 15 degrees from vertical.
VMRS 34-004-021	Lamp, Marker. - All running lamps (clearance, identification, and side marker) per SAE J592, DOT FMVSS, and MIL-STD-1179. - For service, provide 11 each per MS35423 to outline perimeter; 2 front clearance, 4 side marker, and 5 rear clearance and rear identification.
VMRS 34-004-031	Box, Junction. - Connect trailer ends of cable into weatherproof covered, drawbar mounted, electrical junction box having nut and stud type terminal block and having SAE J553 circuit breakers protecting all live trailer circuits. - Connect plug; junction box, and lamps as follows in Tabulation A:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:11Z
Check the source to verify that this is the current version before use.

Tabulation A. Military electrical system connections.

Function	Military Plug Terminals	Connection Circuit Numbers
Blackout Marker	A	24
Service Left Turn and Stop	B	22 - 461
Blackout Marker	C	24
Ground	D	90
Service Tail and Marker	E	21 - 489
Blackout Stop	F	23
Blackout Marker	H	490
Service Right Turn and Stop	J	22 - 460
Auxiliary Power (Hot)	K	37
Ground	L	90
Electric Brake	M	53
Electric Brake	N	53

VMRS No.	Feature/Requirements
VMRS 50	Accessories Group.
VMRS 53-004-013	Chock, Wheel.

- Provide 2 chock blocks and 2 cradles for holding the trailer parked.
- Provide chock blocks per figure 8 or, optionally, MS52127.
- Provide cradles per figure 9. Chock blocks shall withstand servicing unit rollover without damage.
- Provide captive chains of a length to place blocks both ahead of and behind the tires without being taut.
- Provide cradles installed on a vertical flat surface with ears up and the lower loop down having excess chain figure-8-wound over the ears.

VMRS 53-006 Box, Mounted, Tool.

- Provide a second (see 3.10.4) metal or fiberglass, weather-resistant toolbox on the servicing unit to stow and secure parts and tools strictly for the trailer.
- Provide toolbox warning sign (see 3.15.3) to read, "Toolbox for trailer only. Do not use for any life support equipment tools."
- Provide continuous hinge on lid or door having minimum 1/4-inch diameter brass or stainless steel pin.
- Provide tension-type fasteners with strikers to retain lid or door closed.
- Provide hasp to accommodate minimum 3/8-inch padlock shackle.
- Provide minimum 2 each drain holes with elastomer plugs in bottom of box.
- When closed, the lid shall prevent rain and dirt from entering the toolbox. The lid shall open through an arc of at least 90 degrees and shall remain open by its own weight or by holding devices.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:11Z
Check the source to verify that this is the current version before use.

TABLE I. Trailer specifications - continued.

VMRS No.	Feature/Requirements
VMRS 53-999-008	Fluid, Hydraulic Brakes. - Service the vehicle's hydraulically actuated components, which normally use nonpetroleum based, glycol brake fluid as a medium, with DOT FMVSS No. 116, DOT 5 type fluid meeting the minimum requirements of MIl-B-46176 for diorgano polysiloxane fluid (silicones).
VMRS 57	Spare Wheel Mounting.
VMRS 57-001-001	Carrier, Spare Wheel. - Provide a spare wheel mount, complete with an inflated spare tire and wheel assembly of the same size as specified for the road wheels (VMRS 17-001, etc.).
VMRS 57-001-004	Lock, Spare Wheel. - Provide locking means acceptable for a minimum 3/8-inch padlock to prevent loss of the spare tire.
VMRS 57-001-005	Bracket, Spare Tire Mounting. - Shall be mounted on top of the cylinder rack but secured to the trailer frame with a threaded locking device. - Folddown spare tore brackets to include safety features legal in California, if furnished.
VMRS 59	Vehicle Coupling System.
VMRS 59-003	Tow Hooks. - Provide 2 each tow hook and safety chain assemblies per DOT FMVSS [49CFR393.70(f)] and SAE J697. - Provide safety type chain hook on truck ends of chains. - Install so that safety chains extend a minimum 30 inches forward of the center of the tow eye, toward the towing truck.
VMRS 59-004	Drawbar. - Provide A-frame or pole type trailer drawbar on front of trailer. - Provide minimum 72-inch drawbar length, measured form center of tow eye to No. 1 trailer crossmember. - Provide A-frame angle to permit towing truck of 96-inch width to turn minimum 60-degree right and minimum 60-degree left, with respect to the trailer, without interference.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:11Z
Check the source to verify that this is the current version before use.

TABLE I. Trailer specifications - continued.

VMRS No.	Feature/Requirements
VMRS 59-007	Assembly. - Provide tow eye per SAE J847. - Provide tow eye of alloy steel having a case-hardened wear surface for at least 120 degrees on the forward, top, and bottom of the eye. - Locate the center of the tow eye within a maximum tolerance of 1/4 inch from the longitudinal centerline of the trailer. With unit nearly level, provide multi-position vertical range of two eye adjustment from at least 26-inch height per SAE J749, for crawler tractor drawbars, through at least 33-inch height per SAE J849, for truck and trailer pintle hooks. - Provide not less than 4 positions. - Provide change of height adjustment by a mechanic using only common mechanic's tools per 3.4.
VMRS 72-001-004	Bumper Assembly, Complete. - Provide a rear bumper assembly protecting the full width of the rear of the trailer body, equipment, and chassis. - Provide bumper which shall be minimum 4 inches tall. - Provide underneath metal skid plate which functions to prevent the bumper from hanging on rocks and stumps or bulldozing dirt while crossing highway V-ditches.
VMRS 77	Trailer Frame and Support.
VMRS 77-003	Leg Assembly. - Provide front support leg of heavy duty, single leg type having hand crank and holder, bevel gear box, antifriction travel screw, telescopic leg cover fully enclosed, and self-aligning sand shoe. - Provide support leg of either telescopic multi-stage type with minimum 20-inch retraction travel range or folding type trunnion mount type having minimum 14-inch travel range and then swing-up. - Provide clearance for V-ditches without hang-up and minimum 12-inch road clearance with the tow eye centered at maximum pintle height (see SAE J849). - Provide maximum hand crank effort per "Rating Criteria Efficiency" of TIMA RP No. 4 with servicing unit at GVWR. Minimum rated front leg capacity shall be 60 percent of total trailer's GVWR on the VIN plate. - Provide all-weather type, silicone base lubricant in leg mechanism.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:11Z
Check the source to verify that this is the current version before use.

TABLE I. Trailer specifications - continued.

VMRS No.	Feature/Requirements
VMRS 77-003-004	Sand Shoe.
	- Provide equalizing sand shoe having turned-up edges. - Provide minimum 64-square inch projected area.
VMRS 77-012-002	Frame Assembly, Complete.
	- Provide frame assembly per DOT FMVSS, 49CFR393.71(h), with drawbar and tow eye. - Provide frame of all-welded construction with main longitudinal frame rail members of the continuous type from front tow eye to rearmost crossmember. - Provide frame material of at least ASTM A242 or SAE J410 HSLA steel having minimum 50,000 psi guaranteed yield point. No permanent deformation of frame with 175 percent of rated payload (GVWR minus curb weight) applied to the servicing unit equipment mount crossmembers: (1) when supported at tow eye and suspension mounts on frame, and (2) when supported at the tow eye and the rear jack stands (for operating equipment leveling).
VMRS 78	Trim and Miscellaneous Hardware.
VMRS 78-002 VMRS 78-009	Fender Assembly, Rear. Deflector Assembly, Gravel.
	- Provide rigid rear fenders and flexible splash guards for trailer rear running gear. - Provide wheel splash protection per SAE J682 and J683. - Provide lightweight glass-reinforced plastic or aluminum rear fenders to withstand minimum 300-pound load at any 1-square foot area at any one time. - Provide splash guards per TIMA RP No. 41 and TMC-ATA RP No. 701, of antisail type and interchangeable with MS51331. - Provide flexible reinforced elastomer having minimum 3000 psi ultimate strength and inherent cold weather resistance. - Provide splash guards having an integral textured splash surface of short blades or pins. - Mount TIMA splash guards to brackets and fenders by metal clamping strips fastened with hex bolts and nuts through the TIMA/TMC-ATA standard hole pattern. Angle each guard 5 to 10 degrees inward toward the center to reduce water mist during rainy weather.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:11Z
Check the source to verify that this is the currentversion before use.

TABLE I. Trailer specifications - continued.

VMRS No.	Feature/Requirements
VMRS 79	Safety Devices.
VMRS 79-003	Stabilizer, Vehicle, Manual.
	- Furnish a pair of commercial trailer stabilizing jacks, located one on each side prevent tipping backwards when the trailer is parked for equipment operation. - Permanently fasten stabilizing jacks to frame. - Furnish jacks of the type having a swing-down, screw-actuated mechanism and ground contact pads. - Furnish jacks of the type that self store for travel. - Furnish jacks with individual adjustment for side slopes up to 5 percent with trailer level. - Each stabilizing jack shall have a manufacturer's rating of at least 50 percent of the trailer's GVWR on the VIN plate. - Provide maximum crank handle effort per "Rating Criteria Efficiency," TTMA RP No. 4, with servicing unit tires raised clear of the ground. - Jacks shall withstand static load and dynamic equipment operation with trailer supported only by front support and by both rear stabilizing jacks. - When self-stored for travel, locate jacks so as to not decrease the trailer's angle of departure at the rear and not hang up on V-ditches. - Provide chassis levels to permit personnel to jack up and level the operating equipment for level operation on terrain having a slope of up to 5 percent in any direction. - Provide minimum size circular vial bubble levels recessed on each outside corner: 1 each front and 1 each rear for leveling longitudinal axis. 1 each front and 1 each rear for leveling transverse axis. - Levels shall be visible to permit personnel operating the leveling jacks to accurately level the body for operation. - Levels shall be waterproof as installed and recess mounted to prevent damage.

3.11 Mobility. The servicing unit shall conform to the minimum mobility characteristics specified herein. The servicing unit shall be compatible for being towed by military design trucks of the 5/4-ton and larger payload size. The servicing unit shall travel without malfunction, damage, or permanent deformation (see 4.6.2.3) and shall:

a. Travel over smooth primary paved highways at minimum 50 mph speed.
b. Travel over gravel- and dirt-surfaced secondary roads, haul roads, and trails up to a minimum 20 mph speed.
c. Negotiate off-road terrain, grades, side slopes, mud, sand, and snow to and from operational sites.
d. Negotiate hard bottom creek water crossings.
e. Negotiate highway V-ditches without damage (standard Federal Highway Administration ditches, 36 inches wide by 18 inches deep).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:11Z
Check the source to verify that this is the current version before use.

3.11.1 Fording. The servicing unit, without prior preparation, shall ford hard-bottom water crossings to a depth of the loaded rolling radius of the trailer's tires for not less than 5 minutes' duration without damage. After fording, the trailer chassis shall show no evidence of water leakage into waterproof brake components or waterproof electrical lights, and water contamination of the wheel bearing lubricant in the hub shall not exceed a 5-percent increase.

3.11.2 Coupling and jacking. The front support leg shall withstand without damage or permanent deformation, the forces encountered in coupling and uncoupling the servicing unit from prime movers and a free-fall drop of not less than 4 inches onto a concrete surface. From its fully retracted position, the support leg shall raise and lower the servicing unit throughout the full range of travel, both with and without the rear stabilizing stands in place under the rear of the trailer, without evidence of malfunctions, binding, or interference.

3.12 Weights and dimensions. The servicing unit weights and dimensions shall be as specified in table II. The establishing ratings of all servicing unit components and of the vehicle shall not be increased to conform to this specification. The GVWR of the servicing unit shall be in accordance with TTMA RP No. 39 and DOT 49CFR567 and 49CFR568, Certification. The characteristics of the servicing unit shall be when the servicing unit is on a level surface with the inflation of the tires as recommended by the Tire and Rim Association (TRA) for actual gross vehicle weight operation at TRA highway speeds.

TABLE II. Weights and dimensions.

Feature	Requirement
Weights:	
Gross vehicle weight rating (GVWR)	Minimum 4000 pounds
Trailer tow eye weight range, vertically	4 to 7% of unit GVWR
Dimensions per SAE J1100:	
Overall length (OAL) L108	Maximum 180 inches
Overall width (OAW) W103	Maximum 96 inches
Overall height (OAH) H101	Maximum 96 inches
Drawbar length	Minimum 72 inches
Angle of departure with tow eye centered at 30-inch height, H107	Minimum 25 degrees

3.12.1 Vehicle weight classification. The contractor shall make technical data available to the Government to complete a vehicle weight classification form (see 6.6) for the servicing unit. The Government will inform the contractor of the resulting vehicle weight classification numbers. A variable vehicle classification kit in accordance with MIL-S-40626 shall be furnished on each vehicle, with the trailer vehicle class displayed for the actual loaded equipment gross vehicle weight, in the location approved by the contracting officer.

3.13 Lubrication. Only approved lubricants as specified in Department of the Navy NAVSEA 0994-LP-001-9010 shall be used on the life support portions of the servicing unit. Lubrication means shall be provided for all moving parts that

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:11Z
Check the source to verify that this is the current version before use.

require lubrication. The types of lubricants shall be in accordance with, and selected from, SAE J754 and shall be interchangeable with or replaced by military lubricants including those cited in MIL-HDBK-113 (see 6.4). ALl initial fills of lubricants shall be limited to the types approved by the component manufacturers. Lubrication intervals shall be in accordance with SAE J752 as recommended by the component manufacturer. Grease lube fittings shall conform to SAE J534 and shall be of the threaded types. A metallic lubrication chart(s) shall be furnished in accordance with SAE J753 and shall include the interchangeable military lubricants as well as the commercial SAE J754 lubricant designations.

3.13.1 Servicing and adjustment. The contractor shall service and adjust the servicing unit for delivery in operational condition for the user. The contractor shall use his commercial checklist and include at least the following:

a. Adjustment of compressed gas systems, including stowage and securing all plant components.
b. Adjustment of brake system.
c. Electrical system continuity checks and voltage drop check.
d. Torque of wheel lug nuts.
e. Inflation of all tires.
f. Lubrication checks and lubrication of all components with approved products for severe service.

3.14 Treatment and painting. The portions of the servicing unit normally painted, shall be cleaned, treated, and painted in accordance with MIL-T-704, type F or G, and MIL-C-46168. Unless otherwise specified (see 6.2), the finial color shall be green 383 of MIL-C-46168, except for the control panel of the type II servicing unit. The air servicing half panel and the oxygen servicing half of the control panel shall be painted with finish colors black, No. 17038, for air and green, No. 14187, for oxygen; both in accordance with MIL-C-46168. Inaccessible areas of the equipment shall be treated with antirust coating material in accordance with TIMA Technical Bulletin No. 56. After painting, the portions of the equipment fenders exposed to road wheel splash shall be undercoated with MIL-C-62218 according to MIl-R-46164. Wheels, axles, suspension, and moving components shall be kept free of undercoating. Only undercoating compatible with oxygen, as recommended by the undercoating manufacturer, shall be used.

3.14.1 Fungus and moisture resistance. The electrical circuitry, including all components and connection shall be protected from the effects of moisture and fungus growth by an overall treatment with a varnish conforming to MIL-V-173, composition as specified in 3.14.2, with 1 percent copper-8-quinolinolate (by weight) based on the nonvolatile content of the varnish, except:

a. Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.
b. Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a minimum dry-film thickness of 0.001 to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:11Z
Check the source to verify that this is the current version before use.

3.14.2 Composition. Composition I shall be used unless composition II must be used in order to comply with local air pollution regulations in the application of the varnish. When composition II is used, the contractor shall have available evidence, for the Government, that the use of composition II is required and shall certify that the composition II material complies with Rule 442 of the South Coast Air Quality Management District.

3.14.3 Cleaning of life support system. All parts and systems for use with oxygen and compressed air shall be thoroughly cleaned and tested for cleanliness in accordance with the standards of the following publications and procedures: MIL-D-16791, and NEDUINST 71214. Included in the cleaning procedure shall be the requirement to perform an air purity sampling test after the last step of the cleaning procedure is completed. This shall insure that the parts and systems cleaned meet the exposure limit values for breathing gas as stated in NAVSEA 0994-LP-001-9010 chapter 5.

3.15 Identification data plates. Data plates shall be applied to the equipment on metal plates in accordance with the "Name Plate Standards and Practices," as published by the National Association of Name Plate Manufacturers, Inc. (NAME). Identification and shipping data plates shall be of material in accordance with MIL-P-514, composition C. The plates shall be secured to each end item in accordance with the NAME STandards and Practices.

3.15.1 Identification. The equipment shall be identified in accordance with DOT, the manufacturers, and MIL-STD-130 on MIL-P-514, type I, style 1 plates. The data of delivery shall be that of the complete equipment under contract and shall not be prior to date of Government acceptance.

3.15.2 Shipping data. The equipment shall have shipping data applied to MIL-P-514, type III plates. These plates shall show the silhouette of the end items, in transport position, indicating the center of gravity and the equipment manufacturer's approved points capable of being lifted or tied down or, when furnished, the location and capacity of the lifting and the tiedown attachments. The plates shall be attached in a conspicuous protected location on the curbside.

3.15.3 Instruction. The equipment shall be equipped with instruction plates, decals, or diagrams, including warnings and cautions, describing any special or important procedures to be followed in assembling, operating, or servicing. The safety, warning, caution, and notice plates shall be in accordance with ANSI Z35.1 and compatible with the space available. For type II servicing unit, furnish ANSI warnings similar to figure 10 prominently displayed on both front and rear of the panels having dual air and oxygen systems. In accordance with 49CFR177, Hazardous Materials Regulations, the servicing unit shall be placarded with reversible, diamond shaped "NONFLAMMABLE" signs on all four sides. The reverse side of each sign shall be painted the final color specified in 3.14 to blend the sign with the unit and prevent detection of the servicing unit under field conditions.

3.16 Vehicle marking. THe registration numbers and other markings of the equipment shall be in accordance with MIL-STD-642 (see 6.7). The GVWR shall be marked on each forward side of the equipment in a manner to be discernible. The TRA Yearbook's prescribed tire pressures shall be marked on the side of the vehicle or subframe near the tires. Unless otherwise specified (see 6.2), all marking shall be black on equipment painted green.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:11Z
Check the source to verify that this is the current version before use.

3.17 Government-furnished property. When specified (see 6.2), the following property in the quantities indicated will be furnished by the Government (see 6.9):

Item No.	Description	Identification	Quantity for each servicing unit Type I	Type II
1	Cylinder, compressed gas, air, DOT 3AA, 6000 psi	51" to 55" by 9" nominal	8	4
2	Cylinder, compressed gas, oxygen, DOT 3AA, 2265 psi	RR-C-901/1-11	0	4

3.18 Interchangeability. All parts having the same part number shall be functionally and dimensionally interchangeable. Interchangeable parts are defined as two or more like parts possessing such functional and physical characteristics as to be equivalent in performance and durability, and capable of being exchanged one for the other without alteration of the parts themselves or of adjoining parts, except for adjustment, and without selection for fit or performance. When specified (see 6.2), interchangeability between type I and type II panels and parts for the same gases shall be required.

3.19 Noninterchangeability. On type II servicing units, like parts (gages, lines, and fittings) shall not be interchangeable between the air and oxygen systems.

3.20 Workmanship. Defective parts and assemblies which have been repaired or modified to overcome deficiencies shall not be furnished. All workmanship on the equipment shall be in accordance with the engineering and production standards of the compressed gas and equipment trailer industry. The equipment shall withstand the operation specified herein without permanent deformation, breakage of connections, malfunction, or component interference caused by incorrect workmanship. All parts of the equipment, before and after painting, shall be clean and free from sand, dirt, fins, pits, sprues, scale, flux, and other harmful extraneous material. Edges and surfaces exposed to personnel shall be smooth and rounded to the extent that a hazardous surface does not exist.

3.20.1 Castings and forgings. All parts, components, and assemblies of the servicing unit which include castings and forgings shall be clean of harmful extraneous material. Rework shall be limited to procedures which do not reduce mechanical properties or affect function.

3.20.2 Fabrication.

3.20.2.1 Aluminum and steel fabrication. Aluminum and steel shall provide original quality surface finish and shall be free from kinks and sharp bends. Aluminum and steel having eroded surface is not acceptable. The forming of the material shall be done by methods that will not cause damage to the metal. Shearing, punching, and chipping shall be done uniformly, neatly, and accurately. Corners shall be square and true, and all sharp edges and burrs shall be removed. The burned surfaces of flame-cut material shall be ground to remove heat-effected zones, burrs, slag, and sharp edges. Precautions shall be taken to avoid

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:11Z
Check the source to verify that this is the current version before use.

overheating of metal. Heated metal shall be allowed to cool slowly, except where heat treatment is required. All bends shall be made with precise, unyielding dies or jig fixtures to insure uniformity of size and shape.

3.20.2.2 Glass-reinforced plastic fabrication. Glass-reinforced plastic (GRP) shall be smooth surfaced and free from bubbles, glazing cracks, and discontinuities and shall have a uniform GRP thickness. All edges and openings of GRP shall be completely sealed to prevent water intrusion with no evidence of resin-depleted/fiber rich means. The GRP shall be free of parting agents which impair paint adhesion. Welding, bonding, connections, and fabrication of the GRP shall be in accordance with the recommended methods of the manufacturer of the material for this application.

3.20.3 Welding.

3.20.3.1 Aluminum welding. The aluminum surfaces of parts to be welded shall be free from scale, paint, grease, oxide film, and other foreign matter. The welding shall be done by the inert gas, shielded arc method. Welding procedures and repair welding procedures shall be qualified in accordance with ANSI/AWS D1.2 or ASME as given in 3.20.4. Welding shall be in accordance with one of these two documents.

3.20.3.2 Steel welding. The surface or parts to be welding shall be free from rust, scale, paint, grease, mill scale that can be removed by chipping and wire brushing, and other foreign matter. Parent materials, weld filler metals, and fabrication techniques shall be as required to enable the servicing unit to conform to the examination and test requirements specified in section 4. Parts to be joined by fillet welds shall be brought into as close contact as possible and in no event shall be separated by more than 3/16 inch unless appropriate bridging techniques are used. Unless otherwise specified (see 6.2), the welding process used in fabrication of the servicing unit shall be at the option of the contractor. Welding procedures and repair welding procedures shall be qualified in accordance with ANSI/AWS D1.1 or ASME as specified in 3.20.4. Welding shall be in accordance with one of these two documents.

3.20.3.3 General welding practice. Welds may be ground, filed, buffed, or chipped but shall not be hammered or peened and shall be free from discontinuities such as base metal cracks adjacent to or behind welds, cracks in weld metal, undercutting of base metal, lack of fusion with parent metal or between parts, lack of penetration, overlap, abrupt weld face ridges, lack of prescribed fit, or spatter. A weld may be repaired once by rewelding. Any additional rewelding or weld metal deposition shall require prior approval by the Government quality assurance representative. All joint preparation and welding for repair shall comply with the original requirements for the joint. Welding procedures qualification and repair procedures shall be made available for review by the contracting officer or contracting officer's representative.

3.20.4 Welders and welding operators. Before assigning any welder to manual welding work or operator to automatic welding work covered by this specification, the contractor shall obtain certification that the welder or welding operator has passed qualification tests as prescribed by one of the following listed codes for the type of welding operations to be performed and that such qualification is effective as defined by the particular code:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:11Z
Check the source to verify that this is the current version before use.

ANSI/AWS D 1.1, Structural Welding Code, Steel.
ASME Boiler and Pressure Vessel Code, Section IX, Welding and Brazing Qualifications.
ANSI/AWS D 1.2, Structural Welding Code, Aluminum.

Contractors who make only horizontal welds need not qualify welders for "all position welding." The test results shall be made available for review by the contracting officer or the contracting officer's representative. In the event of poor welds, the Government reserves the right to require retesting of any welder or welding operator.

3.20.5 Welding practices.

a. Preheat of materials being welded and maximum interpass temperature during welding shall be in accordance with the supplier's recommendations.
b. Tack welds shall be subject to the same quality requirements as final welds.
c. Work shall be positioned for flat welding whenever practicable.
d. Procedures and sequences shall be such that distortion and shrinkage will be held to a minimum. WHen straightening is required, caution shall be exercised to insure that the straightening process does not weaken the part and the servicing unit.
e. Repair welds shall be subject to the same quality as the specifications call for.

3.20.6 Weld quality.

a. The contractor shall be responsible for determining the inspection of the root pass and final weld to insure that, when the welded parts are assembled together to make the servicing unit, the servicing unit shall conform to the inspection requirements specified in section 4, as a minimum, 100 percent visual inspection of the welds.
b. All weldments shall be free of slag, flux, weld spatter, and other impurities detrimental to either the appearance or strength of the weldment.
c. Undercut in weldments shall be held to a minimum and shall be not more than 1/32 inch deep, or 10 percent of the parent metal thickness, whichever is less.

3.20.7 Brazing. Brazing of steels, copper alloys, and nickel alloys shall conform to MIL-B-7883.

3.20.8 Bolted connections. Boltholes shall be accurately punched or drilled and shall have burrs removed. Integral or separate washers or lockwashers shall be provided. Prevailing torque nuts are preferred in lieu of standard nuts and lockwashers. All capscrews, bolts, washers, and nuts shall be tight and correctly torqued. All fasteners and washers shall be finished wit a corrosion preventative protective coating (including an electrodeposited or mechanical plating or chemical conversion coating).

3.20.8.1 Critical bolted connections. Critical load-carrying bolted connections shall, in addition, be as specified herein. Boltholes shall be reamed as required. All critical junctures and joints shall be fastened with high strength bolts with washers and nuts, hex or 12-point type. Bolts shall

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:11Z
Check the source to verify that this is the current version before use.

conform to IFI Standards 103 and 111, minimum SAE grade 8, ASTM A449. Washers shall be of the flat hardened steel type, sized to spread the clamping loads such that brinnelling or embedding in the component's members shall be prevented. Mating nuts shall be of the prevailing torque type in accordance with IFI Standards 100, 101, 107 and 108, minimum class C, unflanged, or class G, flanged. Fasteners shall be lubricated prior to wrenching. To insure constant and correct torque performance, fastener installation accuracy shall be achieved by the turn-of-the-nut method.

3.20.9 Riveted connections. Rivets shall fill the holes completely. The upset rivet heads shall be full, neatly made, concentric with the rivet holes and in full contact with the surface of the member, and shall be in accordance with SAE J492.

3.20.10 Machining work. All machined parts shall be manufactured to gage through the use of correct jigs and fixtures and shall permit interchangeability of these parts.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspection.

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection comparison (see 4.5).
d. Inspection of packaging (see 4.7).
e. Inspection of transportability (see 4.6.2.6).

4.3 First article inspection.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:11Z
Check the source to verify that this is the currentversion before use.

4.3.1 First article examination. The first article shall be examined as specified in 4.6.1 and 4.6.1.1. Presence of one or more defects shall be cause for rejection.

4.3.2 First article test. The first article servicing unit shall be tested as specified in 4.6.2, table III. All tests shall be performed with the servicing unit trailer mounted.

4.4 Quality conformance inspection.

4.4.1 Examination. Each servicing unit shall be examined for the defects specified in 4.6.1. Presence of one or more defects shall be cause for rejection.

4.4.2 Tests. Each servicing unit shall be subjected to the tests specified in 4.6.2.8. Failure of any test shall be cause for rejection.

4.5 Inspection comparison.

4.5.1 Examination. The Government shall conduct tests/examinations during the production phase of the contract at the discretion or need of the Government any or all of the tests as specified in tests contained herein.

4.6 Inspection procedure.

4.6.1 Examination. The servicing units shall be examined as specified herein for the following characteristics:

Critical

1. Safety device not as specified.
2. Leaks in pressure lines.
3. Any piping not oxygen-clean as specified.
4. Nonconformance to U.S. Navy life support requirements or to DOT FMVSS or FMSCR.

Major

101. Materials not as specified.
102. Materials are not resistant to corrosion or deterioration or treated to be made resistant to corrosion or deterioration n for the applicable storage and operating environment.
103. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.
104. Contractor does not have documentation available for identification of material, material finishes or treatments.
105. Used, rebuilt, or remanufactured components, pieces, or parts incorporated in the servicing units.
106. Maintainability not as specified.
107. Transportability features incompatible.
108. Human factors characteristics not as specified.
109. Safety characteristics not as specified.
110. Environmental features not furnished or not as specified.
111. Servicing unit construction not as specified.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:11Z
Check the source to verify that this is the current version before use.

112. Rack not as specified.
113. Gas flow control system not as specified.
114. Specifications of trailer not as specified.
115. Toolboxes (servicing unit and trailer) not as specified.
116. Military vehicle electrical system not as specified.
117. Weights and dimensions not as specified.
118. Lube and lubrication features not as specified.
119. Servicing and adjustments not made, incomplete, or not as specified.
120. Predelivery checklist not followed as specified.
121. Finish treatment and painting missing, incorrect, or not as specified.
122. Plates, marking, and weight class missing, incorrect, or not as specified.
123. Workmanship not as specified.
124. Welder, welding operator, and weld procedure qualifications not as specified.
125. Welding not as specified.
126. Castings, forgings, and metal fabrication not as specified.
127. Brazing not as specified.
128. Bolted and riveted connections not as specified.
129. Leaks in compressed gas, hydraulic brakes and other lines.
130. Seals do not prevent entrance of foreign material nor prevent leakage of fluids, greases, or air.
131. Assembly incorrect or not complete.
132. Parts or components missing or do not function.

Minor

201. Treatment and painting not as specified.
202. Identification and other required markings not as specified.
203. Departure of workmanship from commercial practice.

4.6.1.1 Certification. The contractor shall insure that the components and materials used in the servicing unit are manufactured, examined, and tested in accordance with the requirements and references specified herein. As a minimum, the contractor and his component sub-suppliers shall be required to satisfy the criteria for the following:

a. In effect at actual time of manufacturer:

(1) NEDUINST 7121A.
(2) DOT FMVSS.
(3) DOT FMCSR.
(4) DOL OSHA Standards and Regulations.

b. In effect at time of solicitation of bid:

(1) NAVSEA INST 10560.2.
(2) NAVSEA 0994-LP-001-9010.
(3) NAVSEA 0910-LP-312-4600.
(4) Industry standardization document compliance.
(5) Comply with the performance requirements specified.
(6) Meet the physical specifications specified.

4.6.2 Tests. Tests shall be in accordance with table III.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:11Z
Check the source to verify that this is the current version before use.

TABLE III. Test schedule.

First Article	Quality Conformance	Test	Test Paragraph	Requirement Paragraph
X	-	Stress Analysis	4.6.2.2	3.10.6, table I VMRS Group 10 VMRS 77-012-002
X	-	Mobility	4.6.2.3	3.11
X	-	Road	4.6.2.3.1	3.8, 3.10.6 & 3.11
X	-	Brakes	4.6.2.3.2	3.8, 3.10.6 & 3.11
X	-	Fording	4.6.2.3.3	3.10.6 & 3.11.1
X	-	Maintainability	4.6.2.4	3.4 thru 3.4.2
X	-	Compressed Gas Operation	4.6.2.5	3.10.3 thru 3.10.3.8.1
X	-	Valve Backpressure Check	4.6.2.5.1	3.10.3 thru 3.10.3.8.1
X	X	Hydrostatic	4.6.2.5.2	3.10.3 thru 3.10.3.8.1
X	X	System Cleanliness	4.6.2.5.3	3.14.3 thru 6.14
-	-	Transportability	4.6.2.6	3.6
See 6.2	-	Environmental	4.6.2.8	3.9 thru 3.9.3
-	X	Production Model Demonstration	4.6.2.9	Section 3

4.6.2.1 Test conditions. The operation and hydrostatic tests shall be conducted after completion of the mobility tests. Prior to testing, lubricate the servicing unit with lubricants as specified in 3.13. Lubricants shall be those designated for use in the ambient temperature at which the test is being performed. Before any road and off-road terrain tests are conducted, measure longitudinal and diagonal points and critical mounting points on the servicing unit. After tests are completed, check the measurements of these points and visually inspect the servicing unit, including all welds and fasteners. During the tests, all service lights and blackout lights shall be illuminated, turn signals operated on turns, and hazard warning operated during stops. The servicing unit tanks shall be full during mobility tests. The apparatus minimum standard of accuracy shall not exceed the following:

Type of Measurement	Percent of Measured Valve
Deceleration	±2
Force	±2
Linear dimension	±1
Speed	±2
Stress	±2.5
Weight	±2

4.6.2.2 Stress analysis. A static stress analysis for the servicing unit shall be calculated containing load, shear, and moment diagrams (see 3.10.6, table I, VMRS Group 10, and VMRS 77-012-002) both while blocked up for equipment operation and while on its running gear. Compare the analysis to the design stress of yield 1.75 times the payload. Demonstrate the ability of the frame to

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:11Z
Check the source to verify that this is the current version before use.

withstand , without permanent deformation, the operating stress, while jacked up on stands, and the transported stress, while being towed on the running gear. Begin by determining the payload, that amount of weight allowance for the operating equipment and accessories and the correct payload distribution, which if necessary for the trailer chassis to weigh in at a value equal to the maximum chassis GVWR cited on the manufacturer's VIN name plate. Further, determine the test weight necessary to achieve 1.75 times this payload. The test weights shall be imposed on the equipment mounting crossmembers. Support the trailer chassis off of the suspension and tires. At first, support the trailer chassis only on the drawbar front tow eye and on the points where the suspension attaches to the chassis frame. Prior to testing, preload the chassis frame to 1.75 times the specified payload to relieve stresses, then unload. Establish and record the dimensional relationship of trailer chassis points at the lowest outside edges of the mainframe members, below the load bed, on both the road side and the curb side as follows:

Point A - Frame height at drawbar-to-bed junction (normally, No. 1 crossmember)
Point B - Frame height just ahead of end of frame
Point C - Frame height midway between points A and B

Impose a test load of 1.75 times the specified payload and record all six points. Unload and record all six points again. Compare the first set of record point measurements with the third set of recorded measurements. Repeat this test with the trailer chassis only supported on the tow eye and the rear operating equipment leveling leg mount points to the chassis frame. If no permanent deformation on any member of the trailer chassis has occurred, the indicated trailer chassis payload rating shall be acceptable.

4.6.2.3 Mobility. Perform all mobility tests (see 3.11) with a fully loaded servicing unit (eight full cylinders, capped, and any other accessory equipment in place on the trailer). At the completion of each mobility phase, examine the servicing unit.

4.6.2.3.1 Road. Tow the servicing unit over roads as specified below. Subject the servicing unit to at least five stops, and coupling and jacking during each phase of the test.

Phase	Course	Mileage	Speed (variable)	Loading
1	Primary highway	75	Up to 55 mph	Loaded
2	Unpaved roads	50	Up to 30 mph	Loaded
3	Off-road terrain (rough terrain not previously leveled or traveled)	50	Up to 20 mph	Loaded

During the road runs and while towing rated payload, perform the following and record actual results:

a. Tow the servicing unit on side slopes up to 10 percent, having the curb side downhill and having the road side downhill. Make full throttle starts and panic stops on the side slope.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:11Z
Check the source to verify that this is the currentversion before use.

b. Tow the servicing unit over a chatter course until at least 2600 (either upward or downward) accelerations of magnitudes between 4 g and 7 g have been applied to a wheel center. The course shall be constructed so that the acceleration magnitudes can be obtained by adjusting the vehicle speed without the frequency of the accelerations exceeding 15 cycles per second (cps) or becoming less than 3 cps. The maximum acceleration magnitude shall not be exceeded, and accelerations having magnitudes less than the minimum shall be disregarded. The course shall be of such length that a wheel center receives, tin one pass over the course, at least 50 consecutive vertical acceleration cycles of any magnitude less than the maximum specified herein. The accelerations shall be measured by a trial test to establish the required speed. Disconnect the recording instruments and complete the test at established speed. The acceleration values used shall be those of the wheel center receiving the highest accelerations. Means for assuring that the test speed is held constant throughout the test shall be provided. Accelerations shall be measured on a running gear member (usually an axle or spindle) that has a motion identical to that of the wheel center.
c. Tow the servicing unit through mud at least 6 inches deep for not less than 100 feet. Any prime mover may be used for this phase.
d. Tow the servicing unit over highway V-ditches (standard Federal Highway Administration type) at least 36 inches wide and 18 inches deep. Negotiate the ditches both perpendicularly and at an angle of 45 degrees to the direction of travel for no fewer than 10 times. Inspect the underside for damage.
e. Measure and record the DOT FMCSR deviation or off-tracking of servicing unit with respect to the prime mover when the servicing unit is towed straight ahead on a paved surface.
f. Jackknife the servicing unit at angles up to 60 degrees from straight ahead of both to the left and to the right.
g. Turn the prime mover, towing the servicing unit, in a rolling sharpest-turning circle for two complete turns and repeat in the opposite direction.
h. When uncoupled from the truck, raise the front of the servicing unit not less than 4 inches from a concrete surface and rapidly allow the servicing unit to fall on the support leg. Inspect and compare before and after measurements for evidence of permanent deformation.

4.6.2.3.2 Brakes. Test the servicing unit for conformance to the DOT FMCSR and FMVSS and to the requirements specified herein. In addition to the DOT legal tests, perform the following:

a. Determine by the drawbar pull method the maximum holding ability of the breakaway emergency brakes. Release the emergency brakes using only tools supplied with each servicing unit.
b. Remove the chock blocks and place them, first, to prevent the servicing unit from rolling forward, and then repeat to prevent the servicing unit from rolling rearward. Do not unsnap the captive chain from the cradle loop.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:11Z
Check the source to verify that this is the current version before use.

4.6.2.3.3 Fording. Tow the servicing unit through creek bed water crossings not less than the depth specified for not less than 2 minutes. Stop the servicing unit in the water and turn off the lights. Turn on the lights and continue through water towing for a total time of not less than 5 minutes. Park the servicing unit out of the water and inspect.

4.6.2.3.4 Failure criteria. Inability of the servicing unit to complete any of the tests specified in 4.6.2.3 through 4.6.2.3.3 or nonconformance to any requirement specified in 3.8, 3.10.6, or 3.11 through 3.11.2 shall constitute failure of these tests.

4.6.2.4 Maintainability. Maintenance operations required during initial production testing shall be accomplished to determine conformance to 3.4. Failure of the support elements to maintain the servicing unit as specified shall constitute failure of this test. Errors or inadequacies in the manuals shall not be considered in assessing the maintenance ratio. The maintenance ratio shall be computed during all first article testing. Nonconformance to 3.4 through 3.4.2 shall constitute failure of this test.

4.6.2.5 Compressed gas operation. Prior to oxygen cleaning, operate and examine the servicing unit to insure the proper operation of the manifold system, valves, and all other components. The unit shall be completely set up for operation using the rear stabilizing jacks, the SCUBA cylinder trough, and other accessory equipment. Charge at least eight SCUBA cylinders, operating all components of the gas flow control system, including cascade valving. Allow the servicing unit to remain pressurized at 3000 psig for 1 hour and check for leaks (soap solution) over the entire piping and gas flow component system. Failure of any component or accessory equipment to function or any evidence of leaks or inability to charge SCUBA cylinders shall constitute failure of this test.

4.6.2.5.1 Valve backpressure check. With all cylinders in an empty state (0 psi) and the cylinder valves, manifold recharging valves, the manifold shutoff valves and quarter-turn ball valves in the closed position, and the manifold filled with distilled water, attach an air source of 8333 psi (5/3 working pressure) for the air system and 5000 psi for the oxygen system to the recharging fitting. Open the recharging valve fully and leave the manifold in this state for 15 minutes. Any increase in pressure in the cylinder, as evidenced by the gage readings or any leakage through the quarter-turn valves, shutoff valves, or any fittings shall constitute failure of this test. All manifolds, one on type I and two on type II, shall require this test. Note: Type II oxygen fitting is different from the air manifold recharging fitting, see 3.1 and 3.10.3.1. Air may be used as a test gas in the oxygen manifold only if the manifold and valves are returned to an oxygen-clean state.

4.6.2.5.2 Hydrostatic. Subject the servicing unit, less cylinders, to a hydrostatic pressure of not less than 8333 psig for the air system and 5000 psig for the oxygen systems. The cylinder leads shall be plugged leak tight. Hold the test pressure for 90 minutes. Any permanent deformation, damage, or evidence of any leakage shall constitute failure of this test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:11Z
Check the source to verify that this is the current version before use.

4.6.2.5.3 Life support system cleanliness. Clean, test, and document the life support system cleanliness as specified in 3.14.3 and 6.14. Inability to clean the system, errors in cleaning procedure, or noncompliance of the system in meeting the exposure limit values for breathing gas as stated in NAVSEA 0994-LP-001-9010, chapter 5.

4.6.2.5.4 Test results. To insure user safety, the contractor shall place the documented test results for hydrostatic (4.6.2.5.2) and system cleanliness (4.6.2.5.3) in the document holder on the inside of the curbside panel door of each servicing unit.

4.6.2.6 Inspection of transportability.

4.6.2.6.1 First article inspection of transportability.

4.6.2.6.1.1 Examination. The first article shall be examined for the defects listed in 4.6.2.7.1. The presence of one or more defects shall be cause for rejection. Any defect(s) shall be corrected and the first article re-examined.

4.6.2.6.1.2 Tests. the first article shall be tested as follows:

4.6.2.6.1.2.1 Tiedown provisions. The tiedown provisions shall be tested in accordance with MIL-STD-209 to prove conformance to 3.6.1. Inability to meet the requirements of 3.6.1 shall constitute failure of this demonstration.

4.6.2.6.1.2.2 Slinging provisions. The slinging provisions shall be tested in accordance with MIL-STD-209 to prove conformance to 3.6.1. Inability to meet the requirements of 3.6.2 shall constitute failure of this demonstration.

4.6.2.6.1.2.3 Rail impact test. The first article servicing unit shall be tested in accordance with the following rail impact test procedure:

Step 1. From one to five railcars shall be positioned on a level section of track. The following conditions shall be satisfied:

a. The total weight of the car(s) shall be at least 250,000 pounds (114,000) kilograms.
b. The couplers between any cars shall be compressed to take up any slack.
c. All the air and hand brakes on the car(s) used shall be set.
d. Any load in or on the car(s) shall be secured to prevent sliding or shifting; any movement greater than 1.97 inches (5 centimeters) shall be justification for retest.
e. The end of the buffer car to be struck must have a standard draft gear.

Step 2. Mount the test item on the test car. The test car shall be equipped with standard draft gears and conventional under frames. Mounting of the test item shall incorporate the standard loading and bracing method as shown in section 6 of the Association of American Railroads (AAR) "Rules Governing the Loading of Department of Defense Material on the Open Top Cars." No exotic or unusual tiedown methods shall be used; any non- standard loading and bracing must be approved by the MTMC/TEA prior to testing. The arrangement of the test item and its blocking and tiedown to be tested shall be identical to that proposed and approved by MTMC/TEA (if nonstandard).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:11Z
Check the source to verify that this is the current version before use.

Step 3. Situate the test car between the buffer cars and the locomotive, and pull the test car at least 200 feet (65 meters) from the buffer car(s) along a level section of track (a minimum distance to achieve the required locomotive speeds).

Step 4. Position the knuckles of the buffer and test cars for coupling.

Step 5. Install a timing device to measure the test car speed (to an accuracy of plus or minus 0.1 km/hr) just prior to impact with the buffer car(s). Suggested methods include electronic timing (microswitches) and radar. he use of torpedoes or a stopwatch is permissible but not recommended because of the inaccuracies involved.

Step 6. Push the test car toward the buffer car(s) and, by using the locomotive's speedometer or other means, release the test car when the desired test speed is reached, thus allowing the test car to freely impact the buffer car(s).

Step 7. Repeat step 6 until the test car(s) has impacted the buffer car three times at the same end, once each at speeds of 4, 6, and 8 mph (6.4, 9.7 and 13 km/hr), + 0.5 mph (0.8 km/hr) or - 0.0 mph (0.0 km/hr). Reverse the test car and repeat the 8 mph (13.0 km/hr) for a total of 4 impacts.

NOTE: Adjustment of the lading or securing mechanism or reconditioning of the bracing or items of securement oar not allowed without a complete retest.

4.6.2.6.1.2.4 Failure criteria. Failure to meet the requirements of 3.6.5 and 4.6.2.6.1.2.3 rail impact test shall constitute failure of this test.

Major

133. Tiedown provisions not as specified (see 3.6.1).
134. Slinging provisions not as specified (see 3.6.2).
135. Air transportability requirements not met (see 3.6.3).
136. Rail transportability requirements not met (see 3.6.4).
137. Highway transportability requirements not met (see 3.6.5).

4.6.2.7 Quality conformance inspection of transportability.

4.6.2.7.1 Examination. Each servicing unit shall be examined for the following defects. The presence of one or more defects shall be cause for rejection.

4.6.2.8 Environmental. When specified (see 6.2), perform the following environmental tests.

4.6.2.8.1 High temperature. The servicing unit shall be stored in an ambient temperature of +155 °F for 4 hours. Care should be taken if compressed gas cylinders are equipped with fusible plug safety devices. Cylinders may be removed at the supplier's discretion while the temperature is above 140 °F, but shall be installed for all other test temperatures. The servicing unit shall then be stabilized at 120 °F. While at the temperature and in an equal ambient

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:11Z
Check the source to verify that this is the current version before use.

temperature, the servicing unit shall be operated as in 4.6.2.5. Malfunction of gages and controlling devices, inaccessibility of controls, damage to any components, or performance not in accordance with 3.9 through 3.9.2 shall constitute failure of this test.

4.6.2.8.2 Low temperature. The servicing unit shall be stored in an ambient temperature of -64 °F for not less than 12 hours and examined for damage. The servicing unit shall then be stabilized at -25 °F and operated as in 4.6.2.5. During the test, all gages and controls shall be checked for proper operation. The accessibility of controls to operators wearing heavy clothes and mittens shall be checked. Malfunctioning of the gages and controls, inaccessibility of controls, damage to any components, or performance not in accordance with 3.9 through 3.9.2 shall constitute failure of this test.

4.6.2.8.3 Precipitation. Park the servicing unit on a horizontal surface and expose to natural or simulated rain. The volume of rain impinging on the horizontal surface shall be not less than 1/4 gallon per minute per square foot. Simulated rain shall originate from a source which produces droplets of uniform distribution and velocity similar to natural rain at the point of contact with the equipment. Expose the top and all four ends and sides to the wind-blown precipitation. After the rain exposure, operate the servicing unit in accordance with 4.6.2.5 for at least 15 minutes. Water leakage in weatherproof areas, malfunction or damage, or nonconformance to 3.9.3 shall constitute failure of this test.

4.6.2.9 Production model demonstration. Tow each production model servicing unit for a minimum of 15 minutes over a course which includes highway and off-road terrain. Demonstrate braking, turning, coupling, and jacking. After the mobility demonstration, perform the compressed gas operation in 4.6.2.5, except the leak check time may be reduced to 15 minutes instead of 60 minutes. Then perform the hydrostatic test in 4.6.2.5.2, except the check time may be reduced to 30 minutes instead of 90 minutes. Malfunction, damage, or nonconformance with section 3 shall constitute failure of this test.

4.6.2.10 Human factors. The human factors engineering characteristics of the servicing unit shall be tested in accordance with the criteria of 3.7 and 3.10.3.2.4.

4.6.2.11 Safety. THe servicing unit shall be tested for conformance with the safety criteria specified in 3.8.

4.7 Inspection of packaging.

4.7.1 Quality conformance inspection of pack.

4.7.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.7.1.2 Examination. Each pack shall be examined for the following defects. Presence of one or more defects shall be cause for rejection. Any defects shall be corrected prior to shipment.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:11Z
Check the source to verify that this is the current version before use.

138. Openings not sealed as specified for level A or C.
139. Protective cover not positioned and secured as specified for level A or C.
140. Tags not attached as specified for level A or C.
141. Marking on tags not applied with waterproof ink as specified for level A or C.
142. Technical publications not preserved as specified for level A or C.
143. Marking missing, illegible, incorrect, or incomplete for level A or C.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or C as specified (see 6.2).

5.1.1 Level A. All external openings into the servicing units shall be sealed with barrier material conforming to MIL-B-121, type I, grade A, class 1. The barrier material shall be secured in place with tape conforming to MIL-T-22085, type II. The protective cover shall be positioned to cover the control panel, then secured in a manner to prevent accidental opening. The servicing unit shall be tagged to indicate that all openings into the unit are sealed and that the seals must be removed prior to putting the servicing unit into service. Tags shall conform to UU-T-81, type A. Printing on tags shall be with waterproof ink. The trailer chassis components shall be preserved and packaged in accordance with MIl-STD-281, level A, for mobile shipment. Technical publications shall be preserved in accordance with MIL-P-116, method IC-1 or IC-3, and placed in the toolbox.

5.1.2 Level C. External openings into the servicing units shall be sealed as specified for level A. Technical publications shall be preserved as specified for level A. Trailer chassis shall require no preservation.

5.2 Packing (see 6.10). Each complete servicing unit, preserved as specified in 5.1, shall be prepared for mobile shipment in accordance with carrier rules and regulations.

5.3 Marking. In addition to any special marking specified in the contract or purchase order (see 6.2) marking shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Each type of servicing unit is intended for refilling SCUBA breathing cylinders in remote field sites. Type I servicing unit cylinders are intended to be replenished with compressed air from special diving air compressors. Type II servicing units are intended to selectively supply compressed air or oxygen. When being replenished, four with compressed air from special diving compressors. Each type of servicing unit is trailer-mounted (balance cart type) and intended to be compatible for towing over highways and off-road terrain to and from various diving sites.

6.2 Acquisition requirements. Acquisition documents shall specify the following:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:11Z
Check the source to verify that this is the current version before use.

a. Title, number, and date of the specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. Type required (see 1.2).
d. Time frame required for submission of first article sample, the quantity of samples, or when a first article sample is not required (see 3.2).
e. When the Government will conduct any or all of the first article model examination and tests. When the Government will conduct some but not all of the first article examination and tests, the contracting officer should specify which examination and tests will be conducted by the Government and which examination and tests will be conducted by the contractor (see 3.2 and 4.1).
f. When air transportability data is required for the Technical Information Package (see 3.6.3).
g. When transportability report required (see 6.2).
h. Cite final color of servicing unit if different than green 383 of MIL-C-46168 (see 3.14).
i. Cite color of vehicle marking if required to be different than specified (see 3.16).
j. When the Government-furnished property and the total quantities needed are required (see 3.11 and 3.17).
k. When interchangeability between type I and type II panels is required (see 3.18).
l. When welding process other than as specified is required (see 3.20.3).
m. When environmental tests are required (see 4.6.2.8).
n. Level of preservation required (see 5.1).

6.3 First article. When a first article inspection is required, the item(s) should be a preproduction model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Lubricants. The contracting officer should furnish a list of military lubricants applicable to the system covered by this specification as contained in Federal Supply Catalog, Department of Defense, Section Identification List C9100-IL for FSC Group 91 (see 3.13).

6.5 Data requirements. The contracting officer should include requirements for such data as the contractor's, and each component contractor's. commercial technical publications, instructional materials, illustrated parts list, maintenance and operation manuals. The technical data should be in accordance with SAE J920.

6.6 Vehicle weight classification. The contracting officer should arrange to furnish the contractor with blank vehicle classification forms and request that these forms be completed and returned to the Government. The technical data thus

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:11Z
Check the source to verify that this is the current version before use.

obtained should be utilized to determine the weight classification numbers for the vehicle. The contractor should then be requested by the contracting officer to apply the vehicle weight classification number on each vehicle of the system (see 3.12.1).

6.7 Registration markings. The contracting officer should furnish a list of the U.S. Army registration markings to be applied to the vehicle of the system (see 3.16).

6.8 Provisioning. THe contracting officer should furnish a list of the U.S Army registration markings to be applied to the vehicles of the system (see 3.16).

"Shipment of the servicing unit shall include repair parts, maintenance tools, operational instructions, and the specified hand tools, accessories and supplies, unless exceptions are provided elsewhere in the contract."

6.9 Government-furnished property. The contacting officer should arrange to furnish the property specified in 3.17, when applicable (see 6.2).

6.10 Packing. Levels of packing have not been included as the packing specified is applicable for all levels. For purposes of packing, level and marking, level A should be used.

6.11 Equipment list. The contracting officer should require that prospective bidders for the life support equipment furnished in the servicing unit must be listed as a supplier on NAVSEA INST 10560.2.

6.12 Transportability report. When specified (see 6.2), the contracting officer should arrange for the contractor to furnish a transportability report in accordance with AR 70-47 and S.O.P 70-38 for the servicing units 95 days prior to delivery of the first article servicing units.

6.13 Guidance. If a conflict arises concerning which procedure or publication to use, the contractor shall notify the Government by letter prior to any work being started.

6.14 Documentation. Documentation for cleaning procedures, air purity sampling test procedures, and results of the first article unit and each production unit shall be furnished in a waterproof plastic holder secured on the inside of the curbside control panel door of each servicing unit. The plastic holder shall be sized to hold at least a 12-month time period of documents.

6.15 Subject term (key word) listing.

Diving equipment
Servicing unit
SCUBA gear

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:11Z
Check the source to verify that this is the current version before use.

6.16 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - ME

Review activity:
Navy - SH

Preparing activity:
Army - ME

Project 3655-A109

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:11Z
Check the source to verify that this is the current version before use.

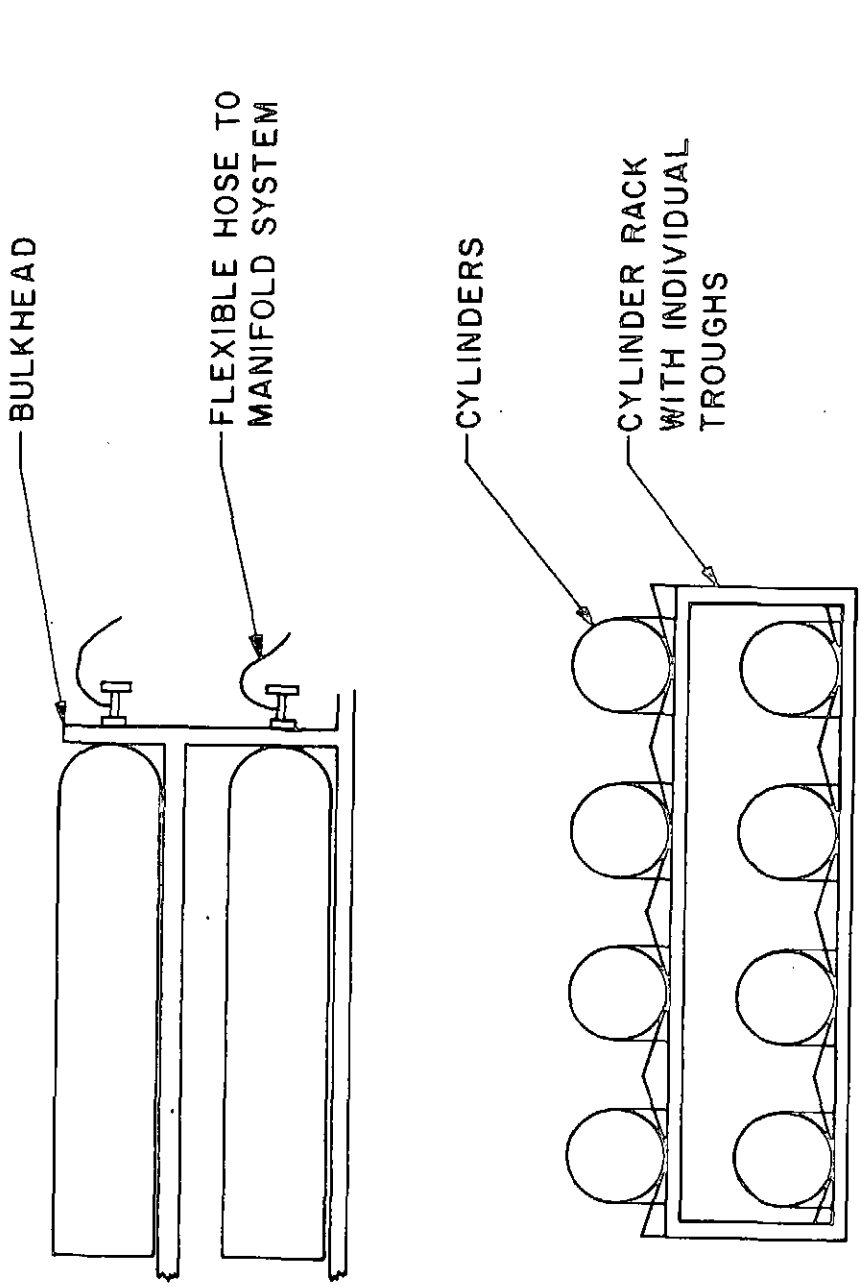


FIGURE 1. CYLINDER RACK TYPE I & II

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:11Z
Check the source to verify that this is the current version before use.

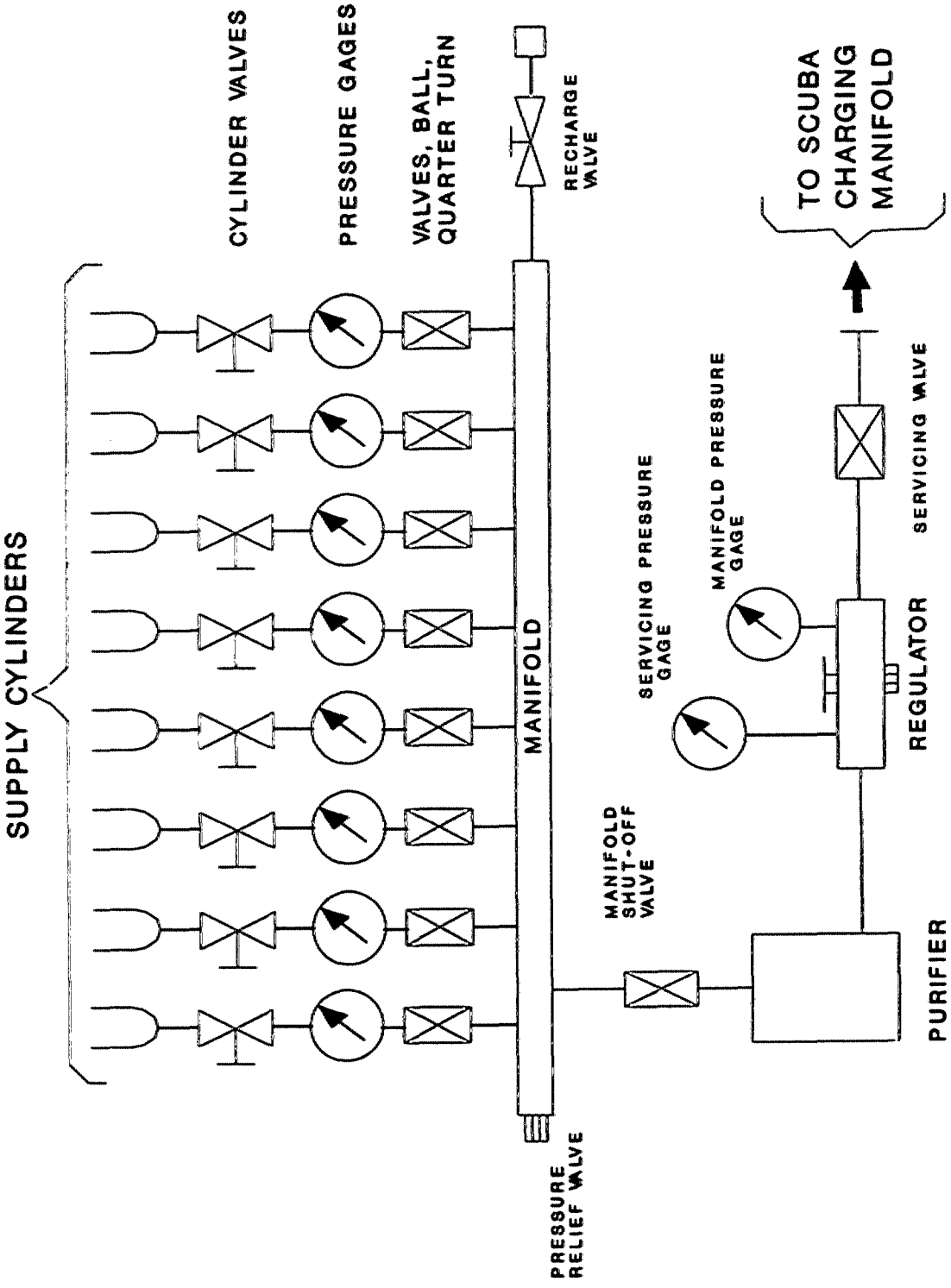


FIGURE 2. Flow diagram, type 1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:11Z
Check the source to verify that this is the current version before use.

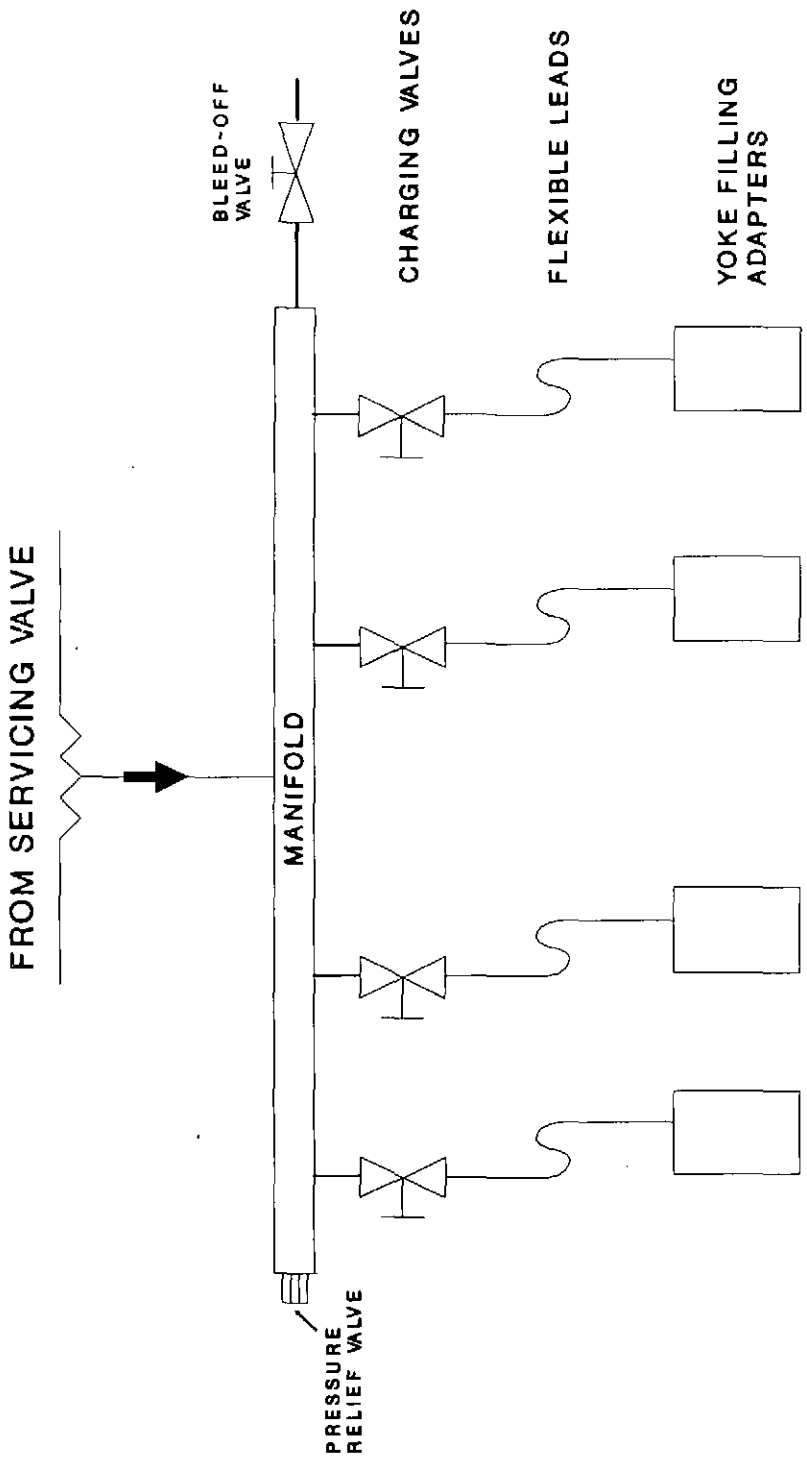
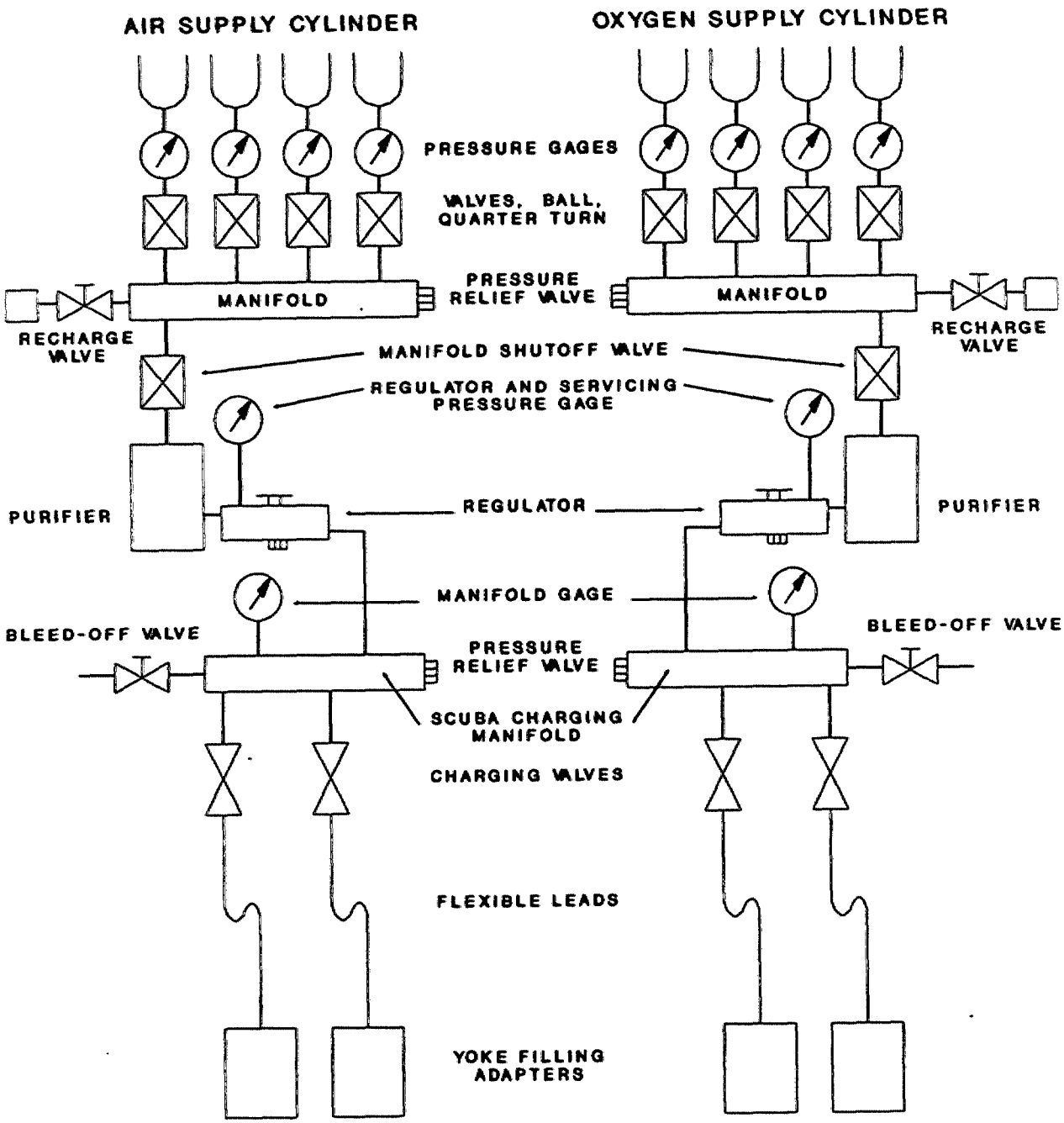


FIGURE 3. SCUBA charging manifold, type I.

X-3028A

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:11Z
Check the source to verify that this is the current version before use.



AIR AND OXYGEN HALVES OF DUAL SYSTEM.

FIGURE 4. Dual air/oxygen gas flow control system, type II.

X-3029B

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:11Z
Check the source to verify that this is the current version before use.

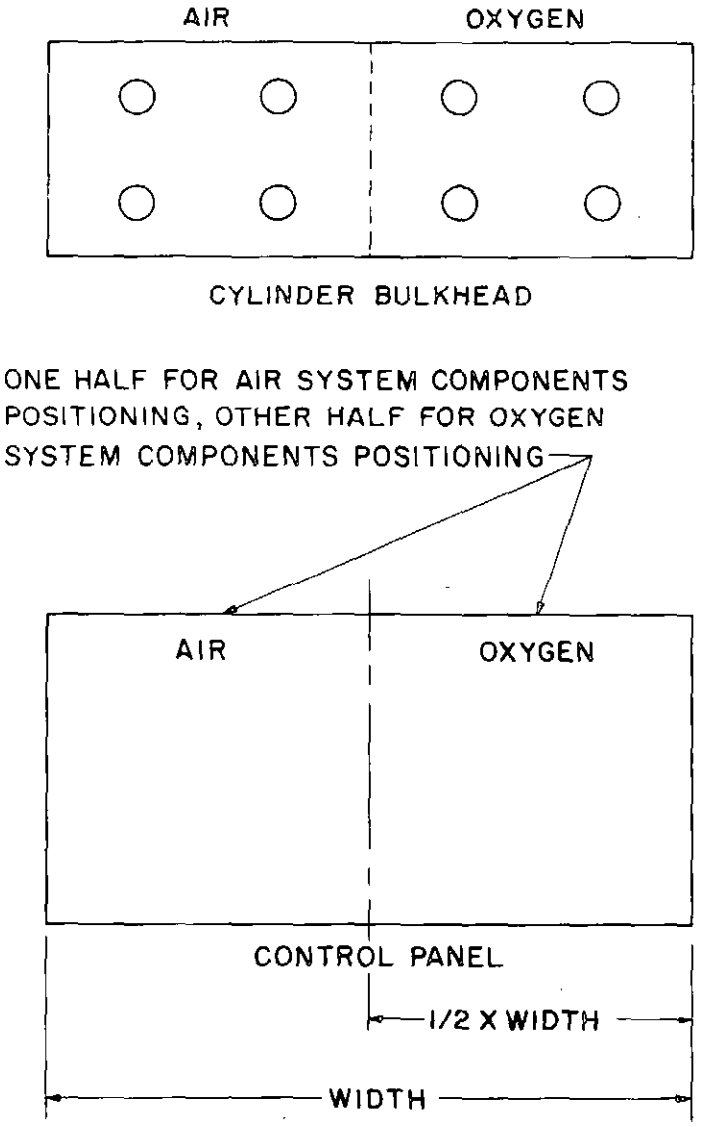
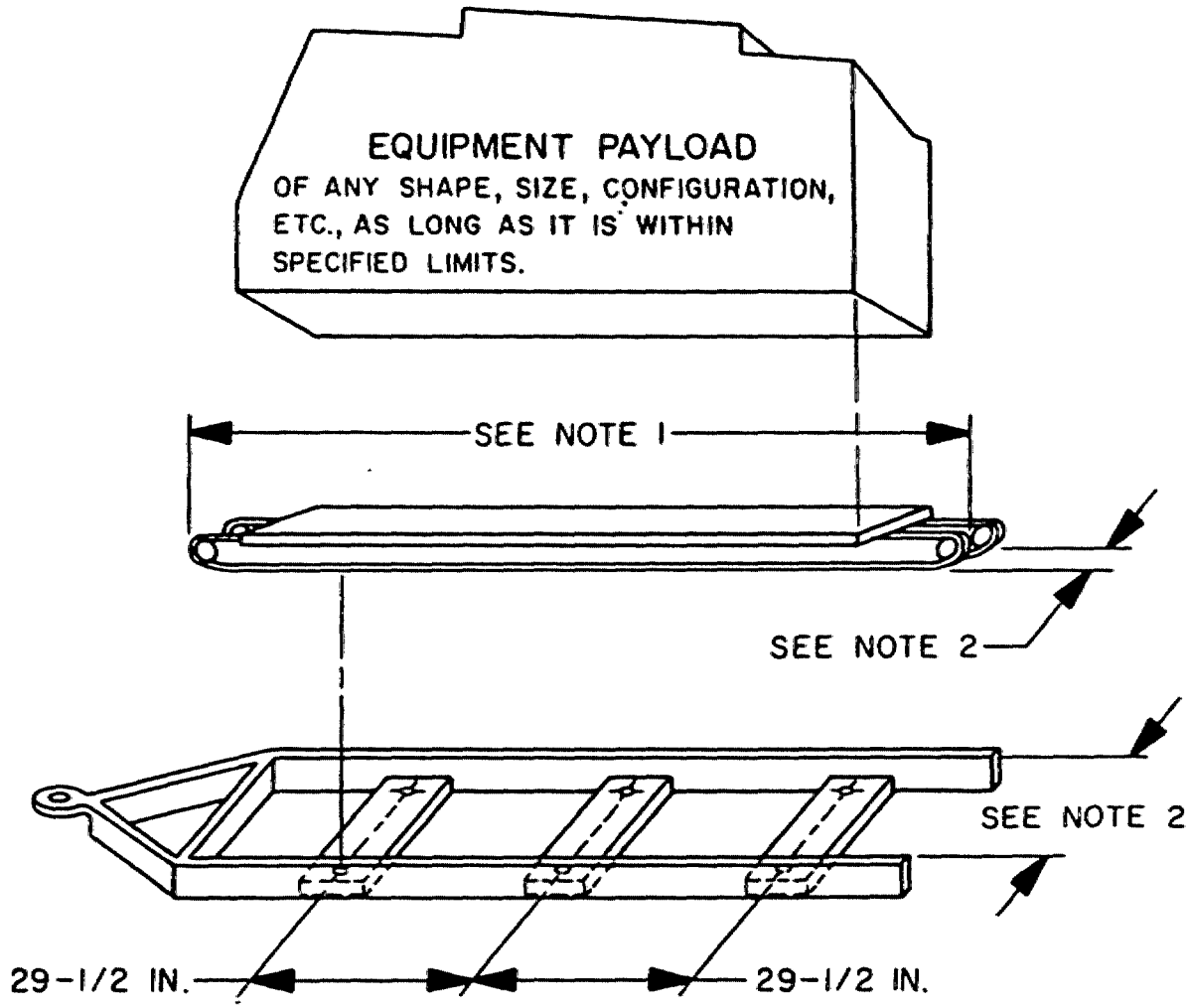


FIGURE 5. HALVING OF DUAL AIR/OXYGEN GAS FLOW CONTROL SYSTEM CONTROL PANEL & CYLINDER BULKHEAD TYPE II

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:11Z
Check the source to verify that this is the current version before use.



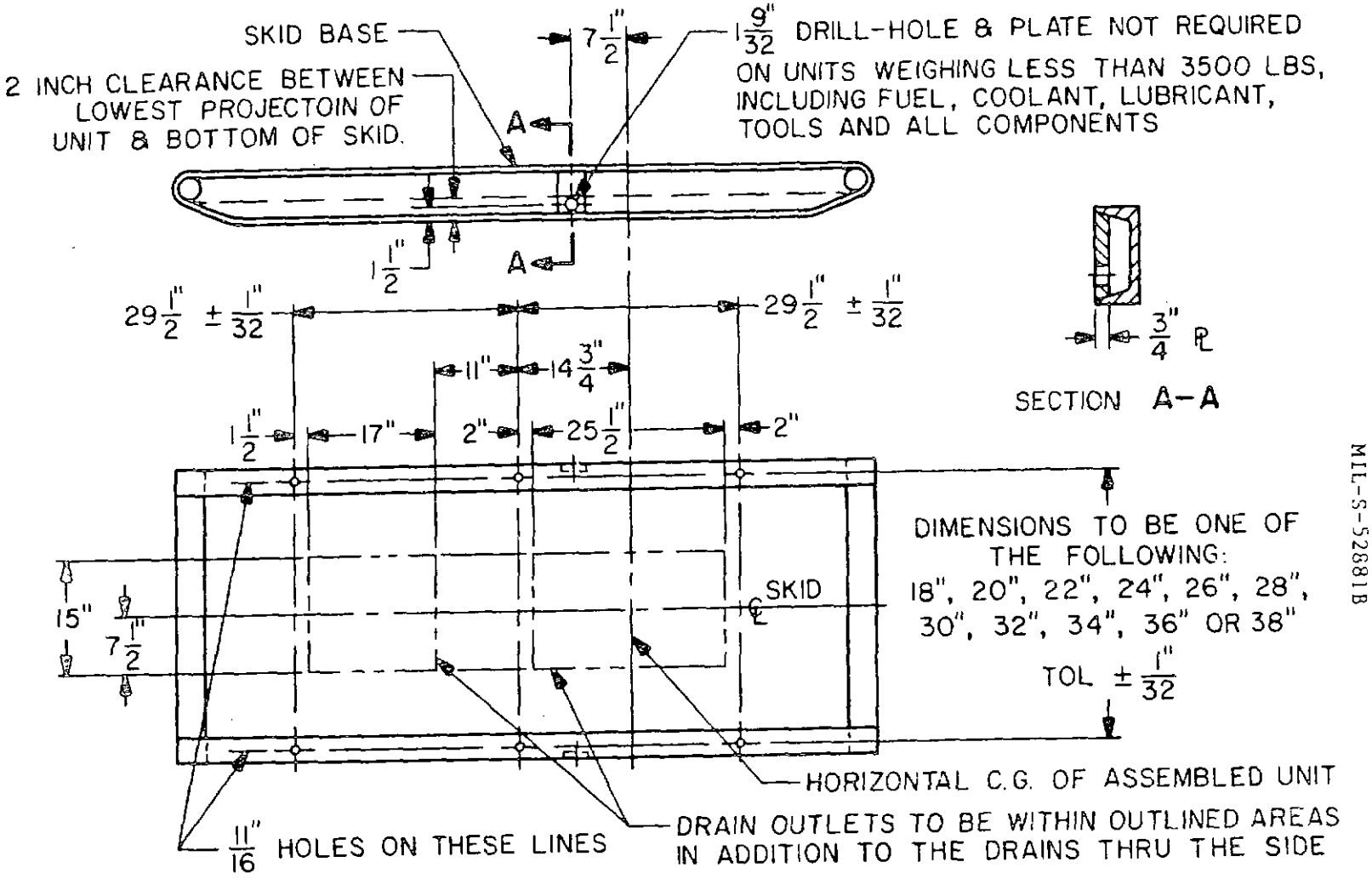
NOTES:

1. SKID DIMENSIONS AS SPECIFIED AND SKID LENGTH OPTIONAL AS LONG AS THE SKID AND THE EQUIPMENT PAYLOAD WILL SLIP WITHIN THE CHASSIS FRAME RAILS AND MOUNT ON THE 3 STANDARD INVERTED CROSS MEMBERS.
2. FABRICATE SKID WIDTH AS NECESSARY TO BOLT DOWN TO THE STANDARDIZED CHASSIS CROSS MEMBERS WITHIN LIMITS.

FIGURE 6. Equipment-to-trailer interface.

X-4842

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:11Z
Check the source to verify that this is the current version before use.



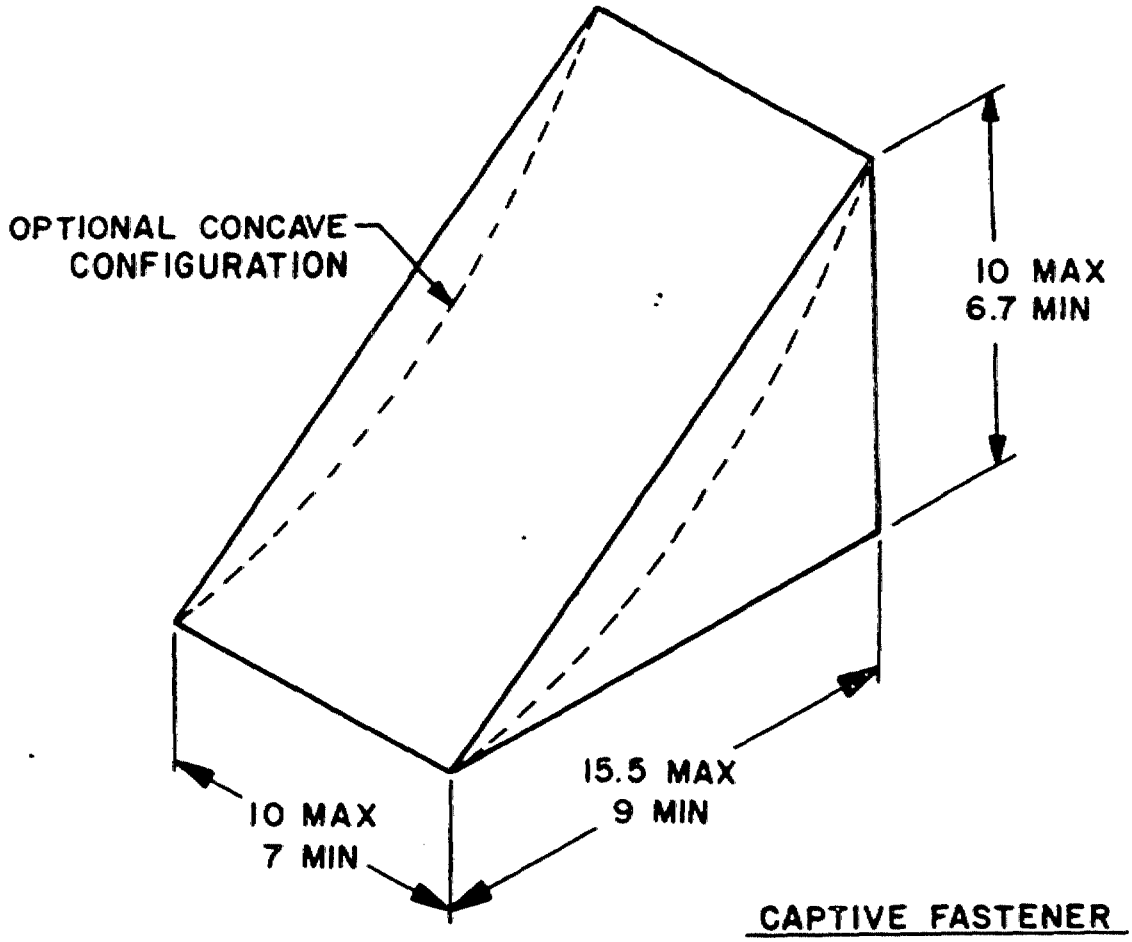
NOTE:

ALL HOLES TO BE LOCATED AFTER DETERMINATION OF HORIZONTAL C.G. OF UNIT INCLUDING FUEL, COOLANT, LUBRICANT, TOOLS AND ALL COMPONENTS.

FIGURE 7. Drilling diagram and drain locations for skid base.

X-4843

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:11Z
Check the source to verify that this is the current version before use.



CHOCK MATERIAL NOTE

1. ALUMINIUM
2. MAGNESIUM
3. SEMI-RIGID ELASTOMER
4. PRESSURE TREATED WOOD
5. MAXIMUM CHOCK WEIGHT-11.5 LB

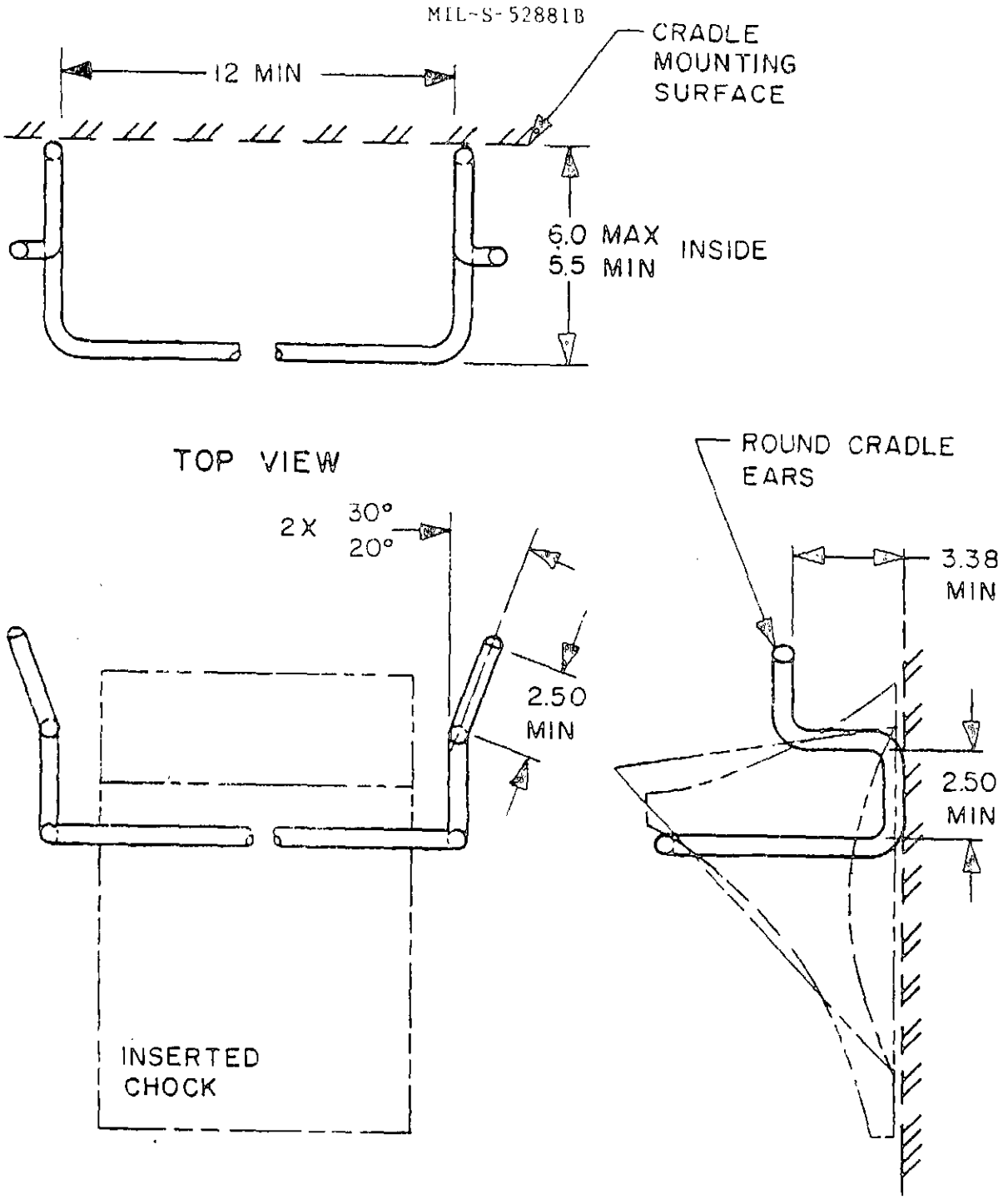
CAPTIVE FASTENER ASSEMBLY DETAILS

1. SNAP HOOK FOR CRADLE OR EYE.
2. FASTENER LINE TENSILE STRENGTH MINIMUM OF 1,400 LB.
3. PLATE FERROUS COMPONENTS TO RESIST CORROSION.
4. LENGTH OF CAPTIVE FASTENER ASSEMBLY SHALL REACH BETWEEN CRADLE OR HANGER AND EMPLACED CHOCK BLOCK AT EXTREME LOCATIONS AHEAD OF AND BEHIND ROAD WHEELS.

FIGURE 8. Chock block assembly.

X-4844

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:11Z
Check the source to verify that this is the current version before use.

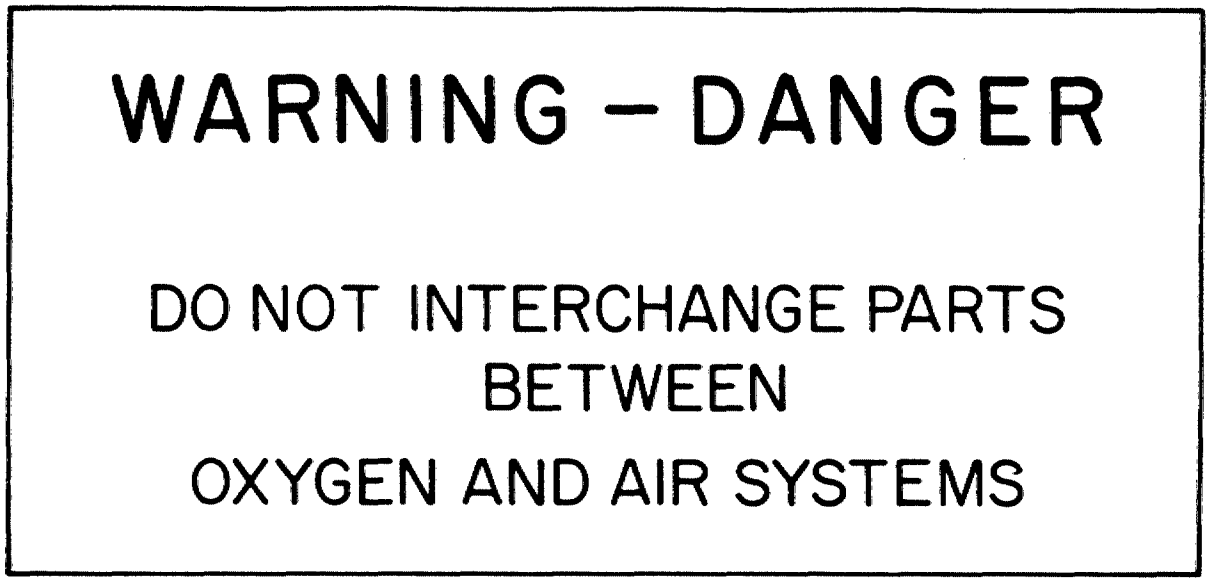


CRADLE MATERIAL NOTE: MINIMUM 1/2 INCH DIA STEEL ROD. BEND RADII 1.0 INCH MIN.

FIGURE 9. Cradle.

X-4845

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:11Z
Check the source to verify that this is the current version before use.



NOTE: LETTERING SHALL BE WHITE ON RED BACKGROUND

FIGURE 1a. WARNING PLATE

X-3031

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:11Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-52881B	Servicing Unit, Compressed Gas, Mobile

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 02 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:11Z
Check the source to verify that this is the current version before use.